

Understanding the perceived human well-being benefits and values of the Palau National Marine Sanctuary: A Nationwide Study



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Glossary of Terms

<i>Geopolitical Terms</i>	
Term	Definition
DFZ	Domestic Fishing Zone, the 20% of Palau's EEZ which is open to commercial fishing by Palauan vessels and foreign vessels by special designation
PNMS	Palau National Marine Sanctuary, the 80% of Palau's EEZ which is fully protected from exploitation
EEZ	Exclusive Economic Zone, the 200 nautical square miles past internal waters of which Palau has full sovereignty over. Both the PNMS and DFZ are contained within the EEZ
<i>Scientific Terms</i>	
Term	Definition
n	Sample size
Population	The full measurement of an identified group
Sample	A smaller segment of the population, selected for the data collection in order to draw conclusions about the population
SE	Standard error, a statistical measurement of precision for a sample, to determine how far it varies from the true population measurement
Spearman's correlation analysis	A non-parametric measurement of relationships between two variables, based on their ranks instead of raw values
Unweighted	All data points have equal value, equal to 1
Weighted	Data points are assigned value based on their relative proportion in the population size

Executive Summary

The purpose of this report is to present the results of the nationwide socioeconomic assessment primarily focused on the Palau National Marine Sanctuary (PNMS). The purpose of conducting this assessment is to understand the community's perception of the PNMS and the benefits they do or do not receive from the PNMS; and to gather suggestions to improve the management of the PNMS.

Key findings:

Support for the PNMS: The study indicated that 70.1% of the respondents showed some level of support for the PNMS: 41.3% “support” and 28.8% “strongly support” the PNMS. A majority of the PNMS supporters associated the PNMS with increasing marine resources, maintaining supplies of seafood, and supporting a healthy ocean.

Changes to the PNMS: A majority of the respondents (56.7%) had not heard of the proposed change to the PNMS (expansion of the domestic fishing zone), while 43.3% had heard of such proposed change. A majority of those who had heard did not support increasing the Domestic Fishing Zone (DFZ).

Fishing activities: This assessment showed that 38.9% of households participated in reef fishing, 2% in pelagic fishing, and 17% in both. The remaining 41.9% of Palauan households did not fish at all. For those engaged in fishing, the main reasons driving them to fish was for household consumption, giving to others, and recreational reasons.

Fish prices: The average cost of pelagic fish reported was \$6.60/pound, of which only one-third of the respondents felt was affordable. In contrast, the average price of reef fish was \$3.20/pound and 78.7% of respondents felt that that price was affordable.

Sources of household income: Notably, although 57.9% of households engaged in some types of fishing, only 2.7% of households listed reef fishing and 0.8% of households listed pelagic fishing as one of their main sources of income. The majority of the main sources of household income were government jobs, government benefits (social security and pension plan), private companies, and “makit” (the informal sale of goods including handmade products, produce, fish, etc). Moreover, of all the households interviewed, annually 21.6% of them earned less than \$10K, 22.8% between \$10K-\$20K, 17.9% between \$20K-\$30K, and 12.7% over \$30K.

Conclusion and Recommendations

This assessment indicated that the majority of Palauan residents showed some level of support for the PNMS. Moreover, significant gaps in knowledge and misperceptions of the PNMS were also identified. The respondents raised significant concerns about the rising cost of and limited accessibility to offshore pelagic fish.

Based on the findings of this assessment, the recommendations are:

- Making offshore pelagic fish more accessible to the fishing communities by installing Fish Aggregating Devices (FADs) in each state.
- Lowering the cost of pelagic fish to make it more affordable to the local community by making pelagic fish more accessible.
- Enhancing community outreach on the PNMS, its boundaries and objectives.
- Conducting follow-up studies to assess the effectiveness of the recommendations above.

Introduction

The Palau National Marine Sanctuary (PNMS) is a large-scale marine protected area (LSMPA) in the exclusive economic zone (EEZ) of the Republic of Palau (Figure 1). It was enacted by Republic of Palau Public Law (RPPL) 09-49 in October 2015, amended in June 2019 with RPPL 10-35, and, after a drawdown period in which foreign commercial fishing reduced operations, entered into full force on January 1, 2020 (Palau National Marine Sanctuary Act, 2015). The PNMS closed 80% of Palau's Exclusive Economic Zone (EEZ), and the remaining 20% remained open as a designated Domestic Fishing Zone, where both domestic and foreign commercial fishing is permitted (Figure 1) (Claassens et al., 2022). The PNMS begins 24 nautical miles away from the shoreline. The first 12 nautical miles are under state jurisdiction and between 12-24 nautical miles are in the Artisanal Fishing Zone (AFZ). The area from 24 to 200 nautical miles lie the boundaries of the Sanctuary, with the exception of the west coast of the main islands which is the Domestic Fishing Zone (DFZ) (PNMS Strategic Plan, 2025).

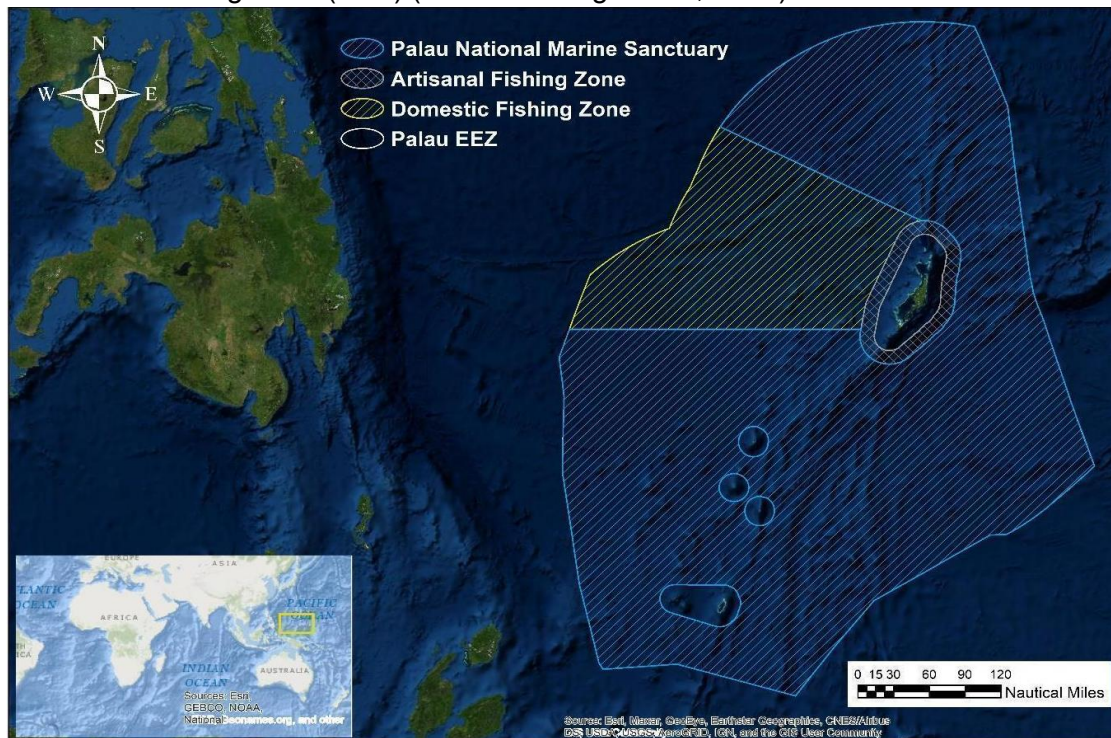


Figure 1. Map of Palau's EEZ with boundaries of the PNMS and designated fishing zones.

Over the past few decades, the implementation of marine protected areas (MPAs), a well-studied natural resource management tool, has progressed from nearshore to offshore areas. Interest has also grown in human well-being dimensions of conservation, prompting efforts to quantify social, cultural, and economic benefits from the ocean, its resources, and the strategies used to manage them. In particular, the remote nature of LSMPAs presents a challenge for imagining human linkages. At the onset of the 2010s, when LSMPAs were proliferating in earnest, their differences in form and function necessitated a separate approach to nearshore MPAs when quantifying social, political, and economic effects to their constituents (Gruby, 2015). For Palau in

particular, a social scientific approach was needed to investigate the human linkages to the Sanctuary.

Thus, a socio-economic study on the PNMS was undertaken by the Palau International Coral Reef Center (PICRC) beginning in 2018, with the implementation of a pilot study in Airai State. Alongside the development of the pilot study, a strategic plan was developed by PICRC to further identify research goals and outcomes for the PNMS (Claassens, 2023). In 2021, that study was expanded to a nationwide project inclusive of all 16 states in Palau. The aims of this study are to 1) understand Palauan residents' perceptions of the PNMS; 2) understand how they believe they do or do not benefit; 3) gather suggestions for improving management directly from the community; and 4) use this information to inform management of the PNMS.

Methods

Project ethics

The human subject research in this study was conducted under International Review Board (IRB) approval granted by the University of Hawaii (Protocol Number 2020-00520).

Site description

The Republic of Palau is an independent country, with an archipelago divided into 16 states (Figure 2). It consists of about 340 islands, eight of which are inhabited, with the population and business center on the islands of Koror. Babeldaob is the largest island by landmass, located to the north of Koror, and is divided into 10 states.

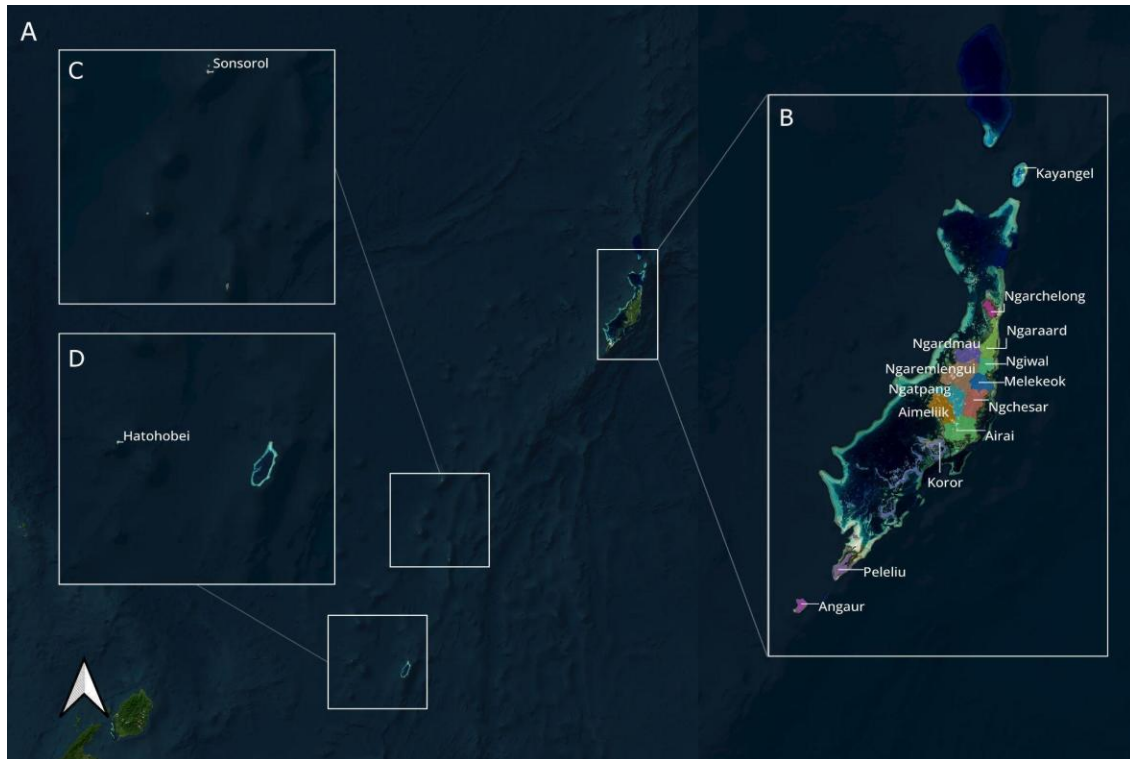


Figure 2. Map of Palau with demarcation of all 16 states. 2A. Map of Palau; 2B. Main Palauan islands (14 states) where the majority of the population resides, overlaid with state names; 2C. Map of Sonsorol State, one of two outer Palauan islands in the Southwest; 2D. Map of Hatohobei State, second of two outer Palauan islands in the Southwest and the southernmost point of Palau.

Household survey development

Household surveys were conducted at all 16 states of Palau, from October 2022 to November 2024. Where possible, community enumerators were deployed with paper surveys, which required no household mapping because of their prior knowledge of communities. For the states where community enumeration was not possible, and for the more populous states of Koror and Airai, PICRC staff enumerated the surveys and used household maps provided by the state, or by the Palau Automated Land and Remote Information System (PALARIS).

The basis for this study's survey tool is human well-being domains which were identified from literature and further refined through focus groups (Koshiba et al., 2022). Focus groups consisted of the following: community men, community women, fishermen, tourism and fishery business owners and operators, and environmental sector stakeholders such as those from non-government organizations (NGO) and government offices dedicated to fisheries and/or conservation. Informant interviews were also conducted to further refine the survey tool. From these processes, six relevant human well-being domains were identified: sustainable livelihood, food security, economic well-being (income), equitable benefits to all, safety and security of life and property, and culture.

A bilingual survey tool (English and Palauan) was developed using these 8 domains. The English tool was pre-tested with 8 households and revised based on feedback before being deployed and was translated into Palauan by a PICRC staff member, Mr. Asap Bukurrou.

In 2022, the survey tool was revised based on feedback from previous enumerators and project staff for relevance and updated research priorities. Questions regarding selling and buying fish, as well as the effects of the COVID-19 pandemic were added into the 2022 survey tool version. Prior to its deployment, the updated survey was piloted with PICRC staff to ensure clarity, measure survey time, and accuracy of the Palauan translations.

Sampling Design

The target population for this study was residential households in all 16 states of Palau. Initial sampling design was a simple random sample (SRS) by state. Household population size was taken from the 2020 Palau Wide Census by PALARIS (Office of Planning and Statistics, 2020).

Households for sampling were identified using household lists provided by each state government. The household lists were separated by hamlet and they were combined to a state level (Koror, Airai, Angaur, Aimeliik, Ngarchelong, Ngatpang). Duplicated, abandoned, unoccupied, or unidentified houses, as well as businesses were removed from the sampling frame. Barracks and apartments were included, and enumerators were instructed to pick one household therein at their discretion. Residences that also served as businesses were included, provided there were residents living there. Rentals were also included as long as they were residential; rentals that house businesses were excluded. Buildings classified as abandoned, under construction, vacant, standalone offices, summerhouses, or demolished were also excluded.

A sample size for each state was then created using a 95% confidence level using Creative Research Systems sample size calculator (<https://www.surveysystem.com/sscalc.htm>). R Studio was used to generate a random, non-repeating number list to randomize households using formula

sample(1:x, y, replace=FALSE)

where **x** is the total household number and **y** is the desired number of households. A sampling frame of additional 10 households was added for each sample size as a buffer to non-responsive or unfit households.

Difficulty in meeting sample sizes occurred due to inconsistencies between census data and state data, and a large volume of households in rural states being uninhabited due to migration, abandoned, or weekend homes, and another SRS was done halfway through the project. Pre-surveys were conducted on states which had not been surveyed to test for similar variables. Due

to time constraints, the survey design was then modified to a nationwide stratified sampling design at 95% confidence level and 5% confidence interval. Post stratification sampling weights are applied for statistical analysis. These weights are applied at the state level for nationwide analysis to adjust for over or under sampling any given state and to ensure that results are reflective of how population is distributed in Palau (Table A1).

The initial goal for the 2022 survey was to sample each state in Palau with enough statistical power for 95% confidence in assessing knowledge, attitudes, perceptions of the PNMS and other livelihood behaviors at the state level. The data collection team was able to sample the following states sufficiently: Aimeliik, Angaur, Hatohobei, Kayangel, and Ngaraard. Due to low sampling power in several states, analysis by region was done to investigate perceptions and behaviors at a more granular level (Table A2). Grouping was done by geographic location and where ambiguous, prioritized population centers (in the case of Ngaraard, which is located in the north of the island of Babeldaob but is populated mostly on the eastern side of the state) (Figure 3). In order to investigate similarities and differences in perceptions of the PNMS across different regions of Palau, states are grouped into geographic regions for further analysis. Additional post-stratification weights were created for analysis of results across the regions (Table A2).

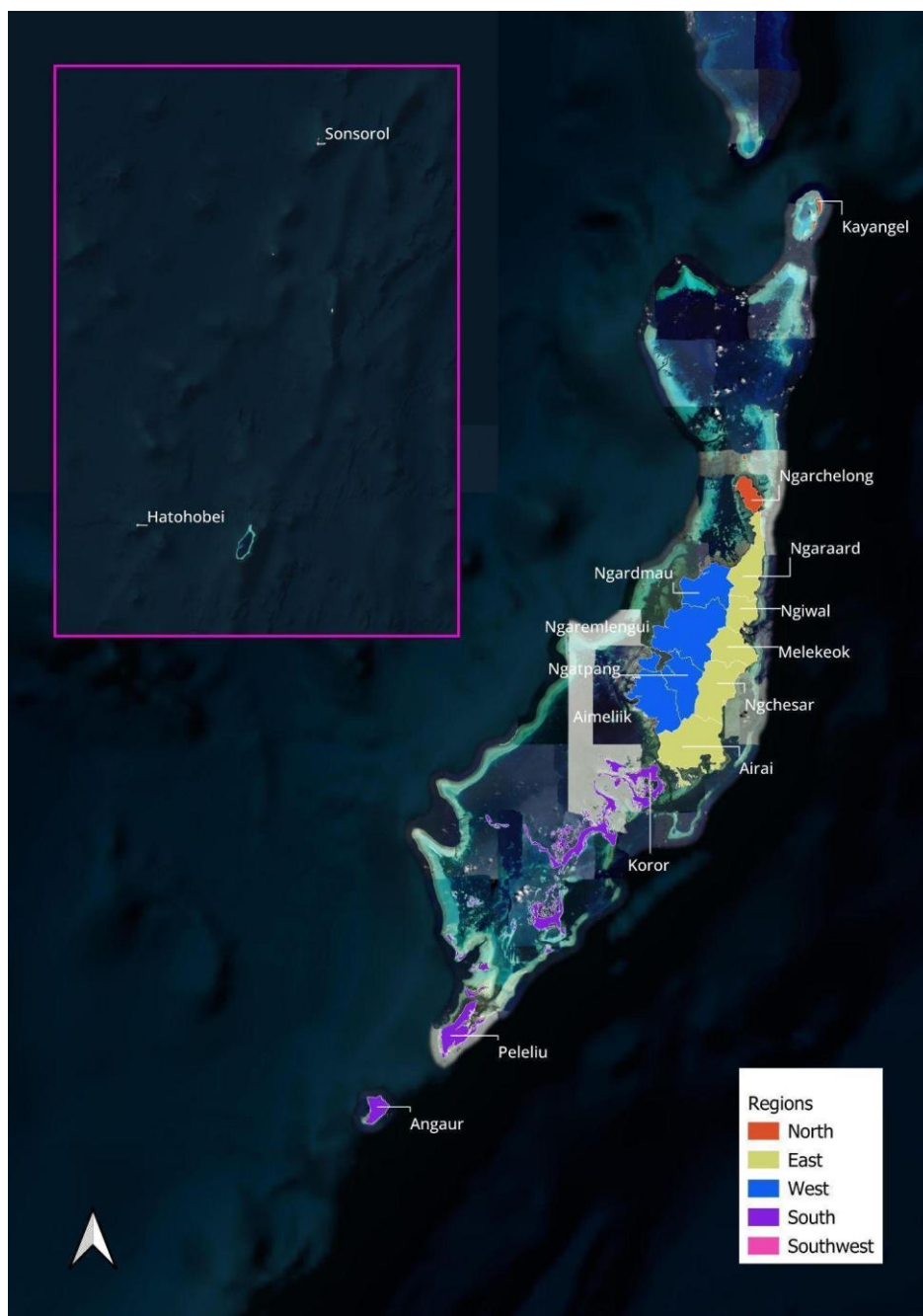


Figure 3. Map of Palau with states grouped by region. The inset image in the map is of the Southwest Island of Palau that is classified into the Southwest region.

Enumeration

Each state initially nominated their own community members to be surveyors, with the number of surveyors dependent on community size. The data collection was supported by independent contractors and PICRC staff. All enumerators received ethics training compliant with rules of human subject research as provided for in the project IRB application.

Data Collection and Analysis

The surveys completed by PICRC enumerators were done directly into the tablets with the Kobo survey tool (<https://www.kobotoolbox.org>). Surveys by community enumerators were first completed on paper and then transferred onto the Kobo survey tool on tablets. All completed questionnaires were uploaded onto the Kobo database for storage and uniform structure, then downloaded and cleaned in Excel. Frequencies and summary statistical analyses were then completed in R Studio using the dplyr package, and analyses with sampling weights were done using the srvyr packages (R Core Team, 2021).

Open-ended answers were analysed using qualitative content and thematic analysis, grouped into categories which presented themselves from the data.

In order to better understand the links between outreach and attitudes, as well as fish preferences and fishing behavior, t-tests were conducted to investigate similarities and differences across pelagic fishing households, reef fishing households, households that use fish in customary or traditional events, and households who fish once per week or more. Spearman correlation analysis was used to identify any correlations between information sources and perceived PNMS benefits. Additional analyses by one-way ANOVA were performed to investigate similarities and differences across the five regions of Palau. Finally, t-tests were used to investigate any changes over time from the first survey in Airai published in 2021. Inferential statistics were completed in PSPP software.

For the regional analysis, post-stratification regional weights were applied to all results. One-way ANOVA analysis with Tukey's HSD post-hoc tests were used to test for similarities and differences across regions. Differences across regions were considered statistically significant at the 95% confidence level. If a question or concept in the survey was not addressed with inferential statistical analysis, there were no statistically significant differences found across the regions.

Survey Questionnaire

Socio-demographic section

Respondents were asked for basic demographic information, such as gender, age, education level, citizenship, household size, and income information, in order to characterize the sampled populations and to draw statistical conclusions among different socio-demographic groups about attitudes and perceptions of the PNMS, as well as fishing behaviors and preferences.

Perceived benefits from PNMS

Statements about the possible benefits of the PNMS were posed to interviewees, who were asked to then rate their agreement to statements from "Strongly Agree" to "Strongly Disagree", "I don't know", or "Unable to answer". They were also asked if they had been negatively impacted by the PNMS, as well as if they felt the needs of their families and communities had been considered in the creation and implementation of the PNMS.

PNMS Knowledge and Support

In order to understand the public support and understanding of the PNMS, interviewees were asked about their level of knowledge, the platform through which they had heard about it, and whether they were aware of specific desired outcomes from the initiative. They were also questioned about their support for the PNMS, their perceptions of equitable benefits, and suggestions for improvement. At the time of the development of the follow-up survey, changes to the PNMS were also being proposed in the Palauan congress. Therefore, respondents were also asked about their support or opposition of changing the law, which will enable better understanding of the link between public support and legislative action.

Sustainable livelihood and food security

One of the aims of the PNMS is to relieve pressure on reef fish by providing pelagic fish as an alternate protein. Respondents were asked about their frequency and category of fishing, fish preferences, and fish sources in order to better understand the local fishery chains. Detailed questions about selling and buying fish, as well as venues of sale and prices, were also included. Responses to questions regarding costs and prices were in US dollars as that is the main currency within Palau. Understanding the exchange of fish within Palauan communities will help to understand if the PNMS has direct or indirect impact on community livelihoods and identify possible areas for intervention and improvement. A question about the 2019 COVID pandemic was also included to gauge if fishing behavior was affected by this global event and to investigate the interplay of the PNMS and the pandemic on fish availability and fishing behavior.

Limitations of the Survey

Sampling Challenges and Adjustments

Although the survey began with a simple random sampling (SRS) design based on the 2020 Palau Census, inconsistencies between census data and state-provided household lists, as well as the presence of a high number of abandoned or uninhabited residences, led to difficulties in reaching target sample sizes in several states. As a result, a second round of SRS was conducted midway through the project. The sampling design was then modified to a nationwide stratified sampling approach, with post-stratification weights applied to improve representativeness.

Incomplete State-Level Coverage

The original goal of achieving statistically significant results at the individual state level (with 95% confidence) was not fully met. Only subsets of states — Aimeliik, Angaur, Hatohobei, Kayangel, and Ngaraard — were sampled with sufficient power for state-level analysis. Consequently, results from other states were aggregated into regional groupings.

Variability in Enumerator Capacity and Training

Enumerator performance varied over the course of data collection. Although all enumerators received training in human subjects research ethics, some state-nominated surveyors were unable to complete the work, requiring PICRC staff to assume data collection responsibilities in the final stages. Differences in enumerator familiarity with the questionnaire and varying levels of experience may have introduced minor inconsistencies in how the survey was administered.

Results

Nationwide Results

Socio-demographics for nationwide sample

For Palau as a whole, the gender split was 50.6% male and 49.4% female. Age distribution was 2% aged 18-24 years, 23.6% aged 25-44 years, 48.4% aged 45-64, 22.7% were 65-79 years old, and 2.9% were over 80. Most respondents indicated some level of formal education, with 35.1% of respondents indicating at least a high school education (Table B1). The majority (91%) were Palauan citizens, followed by 5.2% Filipino, 1.1% Bangladeshi, with all other citizenships being less than 1%. Household size had a mean of 3.67 ± 0.08 , median of 3, and ranged from 1 person to 13 people. The average household reported earning was between \$10,000 and \$19,999 per year (22.8%) (Figure B1). The top two sources of income were government work and government benefits (pension and social security) (Figure B2). “Other” work was the third highest reported income source and was categorized by majority as “private employer”, miscellaneous, and “makit” (the informal sale of goods including handmade products, produce, fish, etc.) (Figure B3).

PNMS knowledge and support

Of interviewees, 43.9% said they had “little level of knowledge” of the PNMS (Table 1). Of those who had at least some level of knowledge, the majority (71.1%) said they heard about it through the internet, social media, or other online sources, followed by newspaper or radio (Figure B4). When asked to speak about the objectives of the PNMS from memory or prior knowledge and understanding, the majority of people connected it to the ocean, with the top three responses being “Increase marine resources” (58.2%), “Maintain supply of seafood” (53.6%), and “Healthy ocean” (52%) (Figure 4). When asked about their level of support for the PNMS, a majority (41.3%) of households said that they supported the PNMS (Figure 5).

Table 1. Level of knowledge nationwide of the Palau National Marine Sanctuary, weighted by state. N (sample size) is denoted in the table in the unweighted count.

PNMS Knowledge Summary			
Category	Weighted %	SE ($\pm$ %)	Unweighted Count
No knowledge	31.4	$\pm <0.1$	492
Little level of knowledge	43.9	$\pm <0.1$	498
Medium level of knowledge	21.1	$\pm <0.1$	231
High level of knowledge	2.5	$\pm <0.1$	38
Extensive level of knowledge	1.1	$\pm <0.1$	17

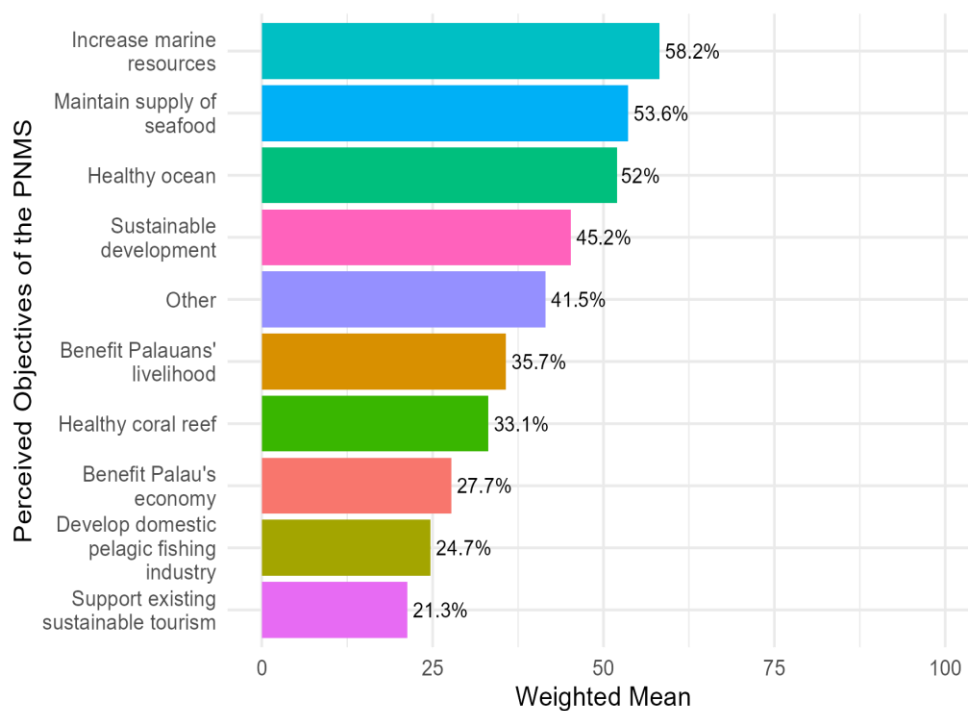


Figure 4. Perceived objectives of the PNMS for respondents nationwide. Standard error is $<0.1\%$ for all.

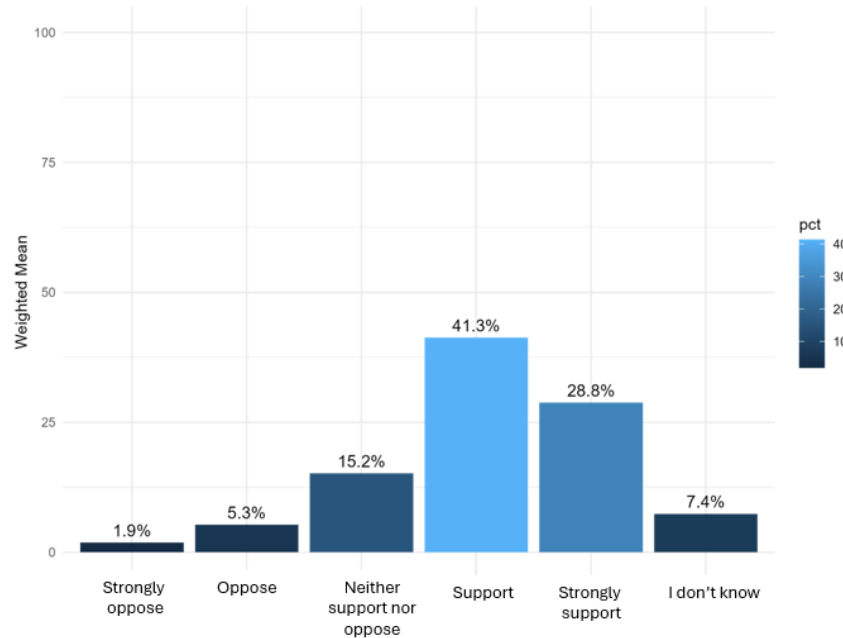


Figure 5. Level of support for the closure of 80% of Palau's Exclusive Economic Zone (EEZ) among respondents nationwide. Standard error is <0.1% for all.

Perceived benefits of the PNMS

When asked if the PNMS had negatively impacted them, 88.4% said that it had not (Table 2). Of those 6.6% who mentioned negative impacts, responded with reasons being “fish had become inaccessible due to price”, “the inability to export fish”, “inability to access fishing in protected areas”, “direct loss of jobs due to the closure of commercial fishing”, and “decrease of flights due to lower tuna exports”.

Table 2. Nationwide weighted responses for negative impacts by the PNMS.

Have you been negatively impacted by the PNMS?			
	Weighted %	SE (±%)	Unweighted Count
No	88.4	± <0.1	440
Yes, specify	6.6	± <0.1	33
I don't know	5.0	± <0.1	23

Respondents were also asked if they had felt considered in the process of creating the PNMS, and the main consensus was that they had been, although more at a community level than a household level. They also felt that food and cultural needs were accommodated at a higher level

than economic or recreational needs (Figure B5). About half of respondents (46.4%) agreed that the PNMS benefits all people of Palau equitably (Figure B8).

Those who answered “Disagree” (6.8%) or “Strongly disagree” (2.3%), said that “those who benefitted the most were those who can afford to fish outside the reef”, “people with money”, and “the government”. They reported those who benefitted the least were Palauans, impoverished people, and those without the equipment to fish outside. When asked who would be negatively impacted, others reported no one, while others reported the community and/or Palauans.

When asked who would be negatively impacted, one respondent said “If we get rid of the fishing industry, you make us very dependent on foreign aid, which is detrimental to Palau. There will be consequences for Palau because we don't have an independent financial situation. By not taking advantage of the resources we have, this hurts all Palauans. We lost a lot of revenue and there were many unintended consequences.” Others reported that the community, locals, and Palauans would be negatively impacted due to the price of fish.

Changes to the PNMS

Regarding legislative changes to the PNMS, 56.7% had not heard of a change, while 43.3% had. Of those who had heard of the changes to the law, 46% did not support a change, 29.5% did support the change, and 24.5% were unable to assess (Figure 6). The reasons from those who opposed the change included: “the law needed more time to take effect for changes to be felt”, “the law was good the way it was”, “not opening it completely but to decrease the protected area size”, and “more information or studies are needed before change”. Those who supported the change stated reasons such as: “supporting a larger domestic fishing zone (the area where Palauans are allowed to fish)”, “the PNMS had contributed to a loss in revenue or that changing the law would help the economy positively”, “support the marine spatial planning process”, or that “more outreach and understanding about the PNMS was required”.

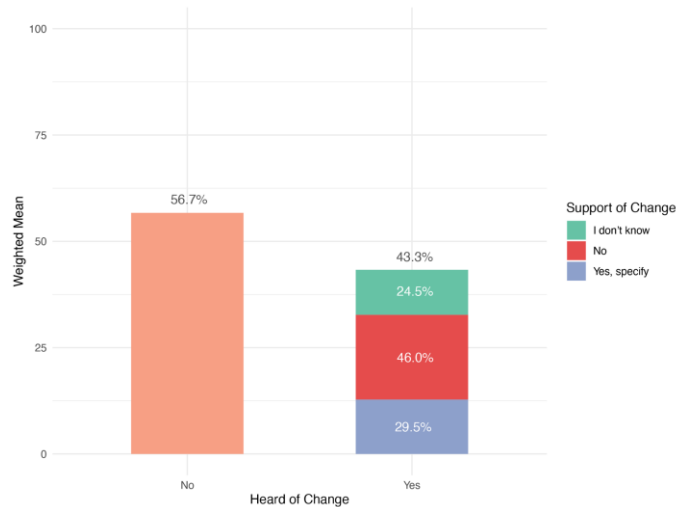


Figure 6. Nationwide awareness of changing the PNMS law. Those who answered “yes” were then asked if they support the change, as displayed in the “yes” bar of the chart. Standard error for all weighted responses was <0.1%.

Understanding perceptions of PNMS benefits

Interviewees were asked to give their level of agreement or opposition to a series of statements regarding the PNMS (Figure B7). Over 80% of the respondents agreed that “the PNMS benefits the next generations.” This statement also received the highest positive answers (“Agree” or “Strongly agree”). On the contrary, the statement with the most negative answers was “the PNMS makes it difficult for my household to fish the way we used to” with over 75% of respondents responding with “Disagree” or “Strongly disagree”.

Sustainable livelihood and food security

When asked about fishing activity in the last 12 months, 38.9% reported they had done reef fishing, 2% reported pelagic fishing, 17% reported doing both reef and pelagic fishing, and 41.9% reported doing no fishing at all. Of reported reef fishers, frequency of reef fishing was mostly one or twice a month (47.1%), rarely (26.6%), about once a week (17.7%), 3-4 times a week (7.4%), or daily (1.3%). Of pelagic fishers, frequency was once or twice a month (41.4%), rarely (43.5%), about once a week (10.4%), 3-4 times a week (4%), or daily (0.6%).

The purposes for fishing is for reef and pelagic fishers are mostly similar, although slight differences were observed. Reef fishers reported fishing mainly for household consumption (96.7%), to give away (85.8%), for recreation (72.6%), for sale (16.5%), and for other reasons (12.2%). Pelagic fishers stated that their main reason for fishing was for household consumption (93.3%) and to give away (92.1%), followed by recreation (69.4%), for sale (23.8%), and for other reasons (19.2%).

Of those who indicated they sold their catch; the majority sold directly to individuals. For pelagic fish sellers, 45.6% reported they sold to individuals, on average at \$2.80 per pound, 35.4% said

they sold to the fish market at an average of \$4.40 per pound, and 31.6% sold to restaurants at an average price of \$5.30 per pound (Table B2). Reef fish sellers indicated that 55.1% sold their catch to individuals at an average of \$2.70 per pound, 39.9% sold their catch directly as “makit” for \$3.40 per pound on average, and 35.5% sold to the fish market at an average price of \$2.70 per pound (Table B3).

Table 3. Businesses, establishments, and people that buy reef fish, and the average price they pay, based on reports from reef fishers nationwide who sell their catch.

Buyer (Reef)	Weighted %	SE (±%)	n (Buyer)	Average Price (\$)	Price SE (±\$)	n (Price)
Community/individuals	55.1	±0.1	97	2.70	±0.41	95
Makit	39.9	±0.1	79	3.40	±0.69	69
Fish Market	35.5	±0.1	59	2.70	±0.21	54
Supermarket	25.9	±0.1	37	2.90	±0.27	37
Caterers	18.0	±<0.1	28	4.20	±1.12	27
Restaurant	15.8	±<0.1	25	2.80	±0.12	21
Hotel Restaurant	13.9	±<0.1	20	3.00	±0.13	19
Others	13.4	±<0.1	19	2.40	±0.12	19
NRFC	5.0	±<0.1	14	2.90	±0.45	14
Government	2.5	±<0.1	7	2.20	±0.25	7
BOFI	0.3	±<0.1	1	NA	±NA	NA

In regard to customary obligations, reef fish was used more often in comparison to pelagic fish. The frequency of using reef fish for customary obligations are as follows: 45.5% never use reef fish, 32.3% once or twice a year, 7.8% every other month, 7.7% once or twice a month, and 6.4% always use reef fish. As for pelagic fish, 60.9% reported never using them, 26.1% use them once or twice a year, 4.8% every other month, 4% once or twice a month, and 3.3% always use pelagic fish for customary obligations.

When asked about fish preference, 50.1% said they prefer both reef and pelagic, 42.6% said they prefer reef fish, 4.7% said they prefer pelagic fish, and 2.5% said they do not prefer either reef or pelagic. The reasons given for preference for pelagic fish were mostly taste (92.4%), followed by it being a high-quality food (87%) (Figure B8). Open-ended reasons given for preference of pelagic fish were mainly to eat it as sashimi; because it could be used in a variety of recipes or cooking styles; that it is healthy; or because it was good to serve to children. The reasons for preference of reef fish were also taste (95.6%), but respondents also highly identified variety (93.8%), knowledge of preparation (91.9%), high quality food (91.8%), and availability (91.1%). Open-ended reasons for reef fish preference were mainly that it was easy to catch or access; it was the food they had grown up eating; that it is healthy; and that it is “fatty.”

Most respondents reported being given fish as their main source (90.9% for reef fish and 85.3% for pelagic fish) (Figure B9). For reef fish, the second most common source was catching it by themselves or within their household (60.9%). For pelagic fish, the second most common source was buying it from the local fish market (48.2%). For pelagic and reef fish, the same open-ended answers were given, including customary obligations, fish markets and co-operatives.

The average price of pelagic fish was \$6.60 per pound, and 33.9% of buyers considered this affordable. For reef fish, the average price was lower at \$3.20 per pound, with 78.2% of buyers finding it affordable (Table 4). Regarding the availability of pelagic fish specifically, 30.4% reported they could sometimes find it for sale, 23.7% said rarely, 19.7% said they did not buy pelagic fish, 8.2% said always, 8% said never, 6.5% said often, and 3.5% gave no response.

The most frequently consumed protein was “other protein” (beef, chicken, or non-fish sources), consumed daily by 16.8%, 3-4 times a week by 37.8%, and about once a week by 26.6% (Figure 7). The most consumed fish was fresh reef fish, which was consumed daily by 10.8%, 3-4 times a week by 26.7%, and about once a week by 26.2%. Fresh tuna was consumed very rarely or never (59.9%), once or twice a month (30.9%), or about once a week (5.9%).

Table 4. Percent of residents nationwide who buy pelagic and reef fish, the mean price they pay per pound, and whether it is affordable or not.

Fish Type	% Who Buy	n	Mean Price/lb (\$)	% Who Say Affordable	n
Pelagic	74.2	792	6.6	33.9	282
Reef	74.2	821	3.2	78.2	659

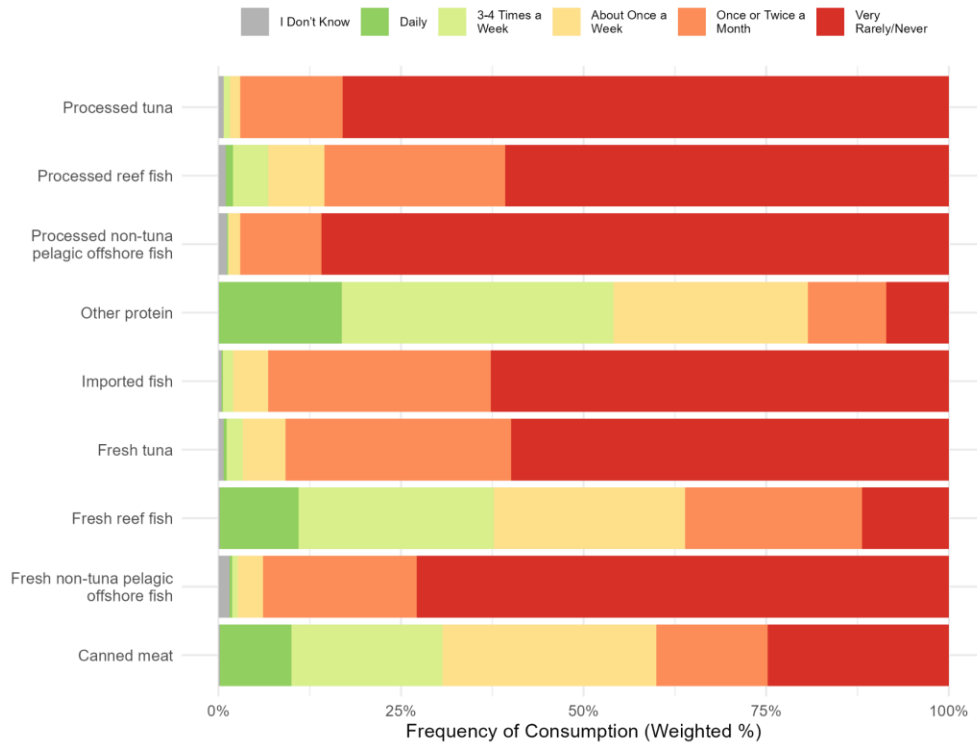


Figure 7. Responses for frequency of consumption of different categories of protein, from respondents nationwide.

When asked about household income sources, very few respondents relied solely on fishing: only 0.8% reported reef fishing as their main income, and none reported pelagic fishing (Figure 8). The most common combination was fishing and government work, with 28.1% of respondents selecting both reef fishing and government work, and 19.8% selecting both pelagic fishing and government work. There was also overlap between fishing types, as 17.6% of pelagic fishers also reported reef fishing while 6.1% of reef fishers also reported pelagic fishing (Figure B10).

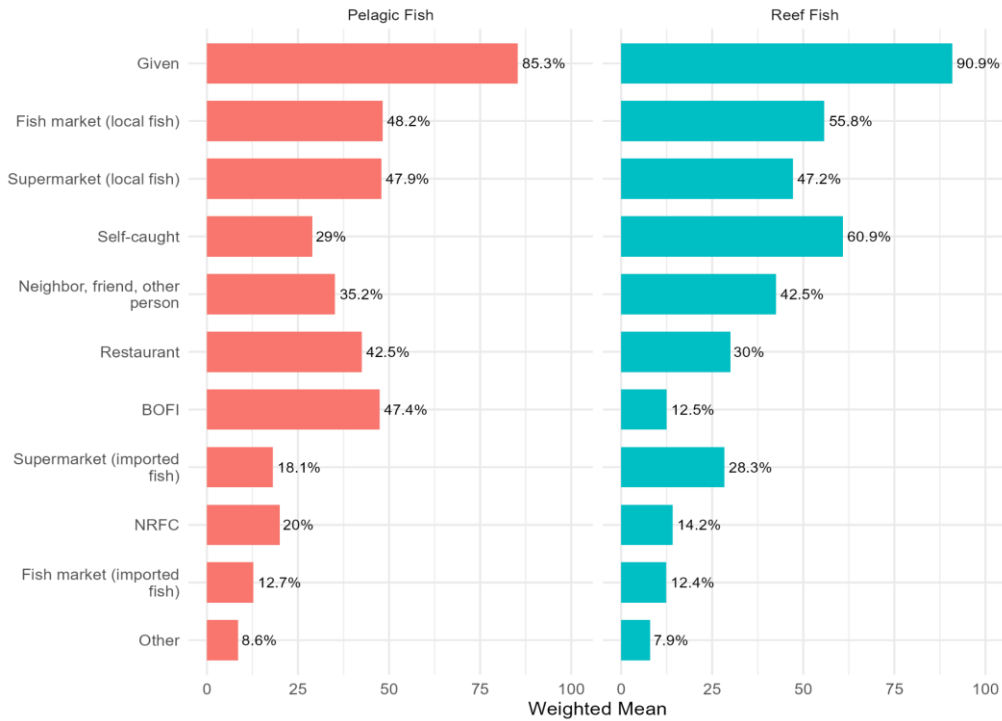


Figure 8. Where respondents source their fish from, as reported nationwide.

Analysis of Nationwide Results Across Questionnaire Categories

This section presents cross-group comparisons of household responses to the PNMS survey, examining differences in knowledge, attitudes, and perceptions based on household fishing practices and cultural use of fish. Sub-groups analyzed include: (1) households engaged in offshore pelagic fishing, (2) households engaged in reef fishing, (3) households using fish for customary or traditional events, and (4) households fishing once a week or more. In addition, a correlation analysis was conducted to assess the relationship between information sources and PNMS perceptions.

All results were weighted at the state level (Table A1), with differences considered statistically significant at the 95% confidence level.

Offshore Pelagic Fishing Household

On average, households that participated in offshore pelagic fishing were more likely to have knowledge about PNMS, be at least moderately knowledgeable about PNMS, and had heard or participated in PNMS outreach when compared to households that did not participate in offshore pelagic fishing (Table C1).

Households that participated in pelagic offshore fishing and had heard of the PNMS were more likely to identify the following PNMS objective correctly: benefit Palau's economy, benefit Palauan

livelihood, develop domestic pelagic fishing industry, and sustainable development when compared to households that do not participate in pelagic offshore fishing and have heard of the PNMS, on average (Table C2).

Answers of “not sure” are coded as missing in Table C3. On average, households that participated in offshore pelagic fishing were more likely to agree that the following needs were taken into consideration during the planning and implementation of the PNMS when compared to households who did not participate in offshore pelagic fishing: family food needs, family economic needs, family recreation needs, family cultural needs, community food needs, community economic needs, and community recreation needs (Table C3).

On average, households that participated in offshore pelagic fishing agreed more with the following statements when compared to households who did not participate in offshore pelagic fishing: “My household has a sustained, safe, and nutritious supply of seafood”, “Offshore pelagic fish is affordable food for my household”, “Permanent closure or no-take from fisheries by PNMS helps Palau sustain and replenish its fisheries”, “PNMS management has a good process to involve communities in their decisions and activities”, or “PNMS makes the conservation efforts of Palau known internationally” (Table 5).

On average, households that participate in offshore pelagic fishing agreed less with the following statements when compared to households who did not participate in offshore pelagic fishing: “Currently PNMS has no benefit for my household”, “PNMS makes it difficult for my household to fish the way we used to”, “Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans”, or “It is necessary to reexamine and revise the way the PNMS law is currently structured” (Table 5).

Table 5. Cross analysis of household agreement to statements regarding the PNMS and participation in pelagic fishing. Post stratification weights by state applied. During analysis, answers of “not sure” were coded as missing, and results that are not statistically significant are not reported.

Please rate your agreement with the following statements	Household does not participate in pelagic offshore fishing		Household participates in pelagic offshore fishing		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
My household has a sustained, safe, and nutritious supply of seafood.	393	3.96	158	4.19	-3.35	<0.01
Offshore pelagic fish is affordable food for my household.	390	2.79	152	3.17	-3.35	<0.01
Currently PNMS has no benefit for my household.	333	2.60	151	2.18	4.76	<0.01
PNMS makes it difficult for my household to fish the way we used to.	351	2.32	149	2.15	2.07	0.04
Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans.	384	2.43	154	2.22	2.02	0.05

Permanent closure or no-take from fisheries by PNMS helps Palau sustain and replenish its fisheries	383	3.99	153	4.19	-2.59	<0.01
PNMS management has a good process to involve communities in their decisions and activities	350	3.67	145	3.87	-2.38	0.02
PNMS makes the conservation efforts of Palau known internationally.	389	4.16	158	4.33	-3.47	<0.01
It is necessary to reexamine and revise the way the PNMS law is currently structured.	316	3.62	123	3.20	3.69	<0.01

Households that participated in offshore pelagic fishing were more likely to indicate that COVID-19 forced them to alter their fishing behavior when compared to households who do not participate in offshore pelagic fishing (Table C4).

Reef Fishing Households

Households that participated in reef fishing were more likely to have any knowledge about the PNMS, be at least moderately knowledgeable about the PNMS, have heard/participated in outreach about the PNMS, and support the establishment of the 80% of Palau's Exclusive Economic Zone (EEZ) as a no-take zone when compared to households that do not participate in reef fishing, on average (Table C5).

Households that participated in reef fishing were more likely to agree that the following needs were taken into consideration during the planning and implementation of the PNMS when compared to households who do not participate in reef fishing, on average: family recreation needs and family cultural needs (Table C6).

Households that participated in reef fishing agreed more with the following statements when compared to households who do not participate in reef fishing, on average: "My household has a sustained, safe, and nutritious supply of seafood" (Table C7).

On average, households that participated in reef fishing agreed less with the following statements when compared to households who did not participate in reef fishing: "PNMS helps the reefs to stay strong to better protect the coastline and coastal communities", "Permanent closure or no-take from fisheries by PNMS helps Palau sustain and replenish its fisheries", or "Changing the current PNMS law will harm Palau's international conservation reputation/standing".

Households that participated in reef fishing were more likely to indicate that COVID-19 caused them to alter their fishing behavior when compared to households who do not participate in reef fishing, on average (Table C8).

Use of Fish for Customary/Traditional Practices

Households that used fish in customary or traditional events were more likely to have any knowledge about the PNMS, be at least moderately knowledgeable about the PNMS, and had heard/participated in outreach about PNMS when compared to households that did not use fish in customary or traditional events, on average (Table C9).

Households that used fish in customary or traditional events were more likely to have heard about the PNMS from the following sources when compared to households that do not use fish in customary or traditional events, on average: factsheets, television, newspaper, radio, and community town hall (Table C10).

Households that used fish in customary or traditional events and have heard of the PNMS were more likely to identify the following PNMS objective correctly: benefit Palau's economy, benefit Palauan livelihood, develop domestic pelagic fishing industry, and sustainable development when compared to households that do not use fish in customary or traditional events and have heard of the PNMS, on average: Benefit Palauan's livelihood, Increase marine resources, Maintain supply of nutritious and safe fish (seafood) for everyone, Develop domestic pelagic fishing industry, and Sustainable Development (Table C11).

Households that used fish in customary or traditional events agreed more with the following statements when compared to households who do not use fish in customary or traditional events, on average: "My household has a sustained, safe, and nutritious supply of seafood", "PNMS helps revive and strengthen our traditional connection to our environment and resources", "PNMS creates the next generation of Palauan leadership to manage Palau's open ocean resources", or "PNMS makes the conservation efforts of Palau known internationally" (Table C12).

Households that used fish in customary or traditional events agreed less with the following statements when compared to households who do not use fish in customary or traditional events, on average: "PNMS makes it difficult for my household to fish the way we used to", "Palau has the needed capacity to manage no-take and domestic fishing zones within the PNMS", or "It is necessary to reexamine and revise the way the PNMS law is currently structured" (Table C12).

Frequent Fishing Households

Households that fished once a week or more were more likely to have any level of knowledge of PNMS when compared to households who do not fish once a week or more, on average (Table C13).

Households that fished once per week or more agreed more with the following statements when compared to households who did not fish once per week or more, on average: "Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans", "It is necessary to reexamine and revise the way the PNMS law is currently

structured”, or “Changing or revising the current PNMS law will bring prosperity and increased economic opportunity to Palau” (Table C14).

Households that fished once per week or more agreed less with the following statements when compared to households who did not once per week or more, on average: “Offshore pelagic fish is affordable food for my household”, “Permanent closure or no-take from fisheries by the PNMS helps Palau sustain and replenish its fisheries”, “Changing the current PNMS law will harm Palau’s environment”, or “ Changing the current PNMS law will harm Palau’s international conservation reputation/standing” (Table C14).

A Spearman correlation analysis was conducted to determine if obtaining information regarding the PNMS from different information sources is correlated with perceptions of the PNMS (Table C15). Each cell contains the Spearman correlation coefficient for the information source and the statement. Correlation results are only shown if they are statistically significant. The range for the coefficients is -1 to 1. Values closer to 1 indicate a stronger positive correlation, and values closer to -1 indicate a stronger negative correlation. Values closer to 0 indicate a weaker correlation. These results indicate that obtaining information from school outreach programs, fact sheets, newspaper, and radio is correlated with more positive perceptions of the PNMS, whereas obtaining information from community town halls is correlated with more negative perceptions of the PNMS. Obtaining information from the internet and symposia is associated with mixed perceptions of the PNMS, both positive and negative.

Regional Results

In this report, we also present regional results as the data did not meet the sufficient power needed to do a state-level analysis. Therefore, the socio demographics of each state were combined into designated regions, and an analysis was done accordingly (Table 6).

Socio-demographics

North

The northern region in this study is composed of the states of Kayangel and Ngarchelong, and the household population in this region makes up 2.58% of Palau’s total household population (Table A2). Respondents were 57.9% male and 42.1% female; primarily Palauan citizenship at 97.9%; ages among respondents in the North were primarily 45-64 years old (38.9%), followed by 65-79 years old (35.8%), 25-44 years old (21.2%), 80 and over at 2.1%, and only 1.1% were 18-24 years old. More than half had at least a high school education, with 53.7% reporting completion of high school, 27.4% having a two-year college degree, and less than 18.9% reporting the remaining categories of elementary school, vocational school, university or graduate school, or no formal education. Income was mainly from government work or government benefits, and the majority earned between \$5,000-19,000 a year, or refused to answer (Figure D1a and D1b).

East

The Eastern region is made up of the states of Ngaraard, Ngiwal, Melekeok, Ngchesar, and Airai, and is 22.60% of Palau's household population (Table A2). Respondents were primarily Palauan citizens at 94%, 3% Filipino, and the remaining 3% were Bangladeshi, American, Chinese, or Micronesian. Interviewees were 51.2% male and 48.6% female; more than half (53.3%) were 45-64 years of age, followed by 21.2% at 65-79 years old, 18.8% were 25-44 years old, 3.4% at 18-24 years old, and 3.3% at 80 years old and over. High school education was reported by 38.5%, followed by 31.7% with a two-year college degree, and less than 23.8% each for the remaining categories of elementary school education, vocational training, university or graduate level degrees, or no formal education. The most commonly reported income bracket was \$10,000-19,000 a year, and the most commonly reported source of income was government work (29.1%) (Figure D2a and D2b).

West

The western region of Palau is characterized as the states of Ngardmau, Ngeremlengui, Ngatpang, and Aimeliik, which together make up 7.13% of the household population in Palau (Table A2). Citizenship is predominantly Palauan (96%), with 2.6% Filipino, 0.9% Bangladeshi, and 0.4% "other" citizenship. Gender distribution was 57.3% male and 42.7% female; the majority of interviewees were 45-64 years old (46.7%), followed by 25-44 years old (26%). The remaining 27.3% was distributed among the other age groups. High school education was reported by 43.6%, followed by 22% with a two-year college degree, and 34.4% for the remaining categories of elementary school education, vocational training, university or graduate level degrees, or no formal education. A clearer majority of households were making \$10,000-19,999 annually, and a third (32.4%) reported government work as an income source (Figure D3a and D3b).

South

The southern region is defined here as the states of Koror, Peleliu, and Angaur, and makes up the majority of Palau's household population at 67.23%, concentrated mainly in Koror (Table A2). Palauan citizenship is the lowest percent of any region at 89.7%, and represents the most diversity with 5.8% Filipino, 1.5% Bangladeshi, 0.6% American, 0.6% Japanese, 0.4% Taiwanese, 0.2% Chinese, and 0.9% "other" citizenship. Respondents were 50.8% male and 49% female; reported age was majority 45-64 years old (48%), followed by 24.9% at 65-79 years old, 23.2% at 25-44 years old, 3% 80 and over, and 0.6% 18-24 years. The majority reported at least a high school education with 34.8%, followed by 29.7% with a two-year college degree, 12.9% obtaining a 4-year university degree, 9.7% an elementary education, and 5.8% a graduate level or higher. Household income level was reported as a majority with 26.2% earning \$10,000-19,999, with 31% reporting government benefits as their top source of income (Figure D4a and D4b).

Southwest

The southwest islands consist of Sonsorol and Hatohobei state, located about 450 kilometers southwest of the main Palauan islands and a 3-day boat ride away (Marino, 2019). They are culturally and linguistically unique to the rest of Palau and part of their population is transient, with

settlements in Sonsorol state (consisting of Sonsorol island, Fanna, and Pulo Anna) and Hatohobei state, and a rangers' station inhabited by 3-4 rangers on 3-month assignments on the southernmost island of Helen (an atoll which hosts Hotsarihie Marine Protected Area, also known as Helen Reef Conservation Area, the southernmost marine protected area in the Palau Protected Areas network, and part of Hatohobei state) (Marino et al., 2019) (Helen Reef Board, n.d.). Many families have settled permanently in Koror, in the village of Echang, a subdivision of Ngerkebesang hamlet in the south of Koror. An open exchange of people and goods is possible due to a regular ship schedule between islands every 3 months.

The states of Sonsorol and Hatohobei make up the southwest region, with 0.46% of the household population of Palau (Table A2). Reported citizenship was 91.3% Palauan and 8.7% Micronesian. Gender distribution was 56.5% male and 43.5% female, with age distribution being the youngest of all regions: the majority of interviewees (52.2%) were 25-44 years old, followed by 34.8% at 45-64 years old, 8.7% 65-79 years old, and 4.3% 18-24 years old. Most were high school educated (65.2%), followed by 26.1% with two years of college, and 4.3% with an elementary school education. The most commonly reported annual income was \$5,000-9,999, and the majority of income sources were reported as government work (63.6%) (Figure D5a and D5b).

Table 6: Summary table of socio demographics within the five different regions. More detailed graphs about education and income sources are found in Appendix D.

	North	West	East	South	Southwest
States Included	Kayangel Ngarchelong	Ngardmau Ngeremlengui Ngatpang Aimeliik	Ngaraard Ngiwal Melkeok Ngchesar Airai	Koror Peleliu Angaur	Hatohobei Sonsorol
% of HH Distribution	2.58%	7.13%	22.60%	67.23%	0.46%
Citizenship	97.9% Palauan	96% Palauan 2.6 % Filipino 0.9% Bangladeshi 0.4% Other	94% Palauan 3% Filipino 3% Other	89.7% Palauan 5.8% Filipino 1.5% Bangladeshi <1% Other	91.3% Palauan 8.7% Micronesian
Age Distribution	45–64 (38.9%) 65–79 (35.8%) 25–44 (21.2%) 80+ (2.1%) 18–24 (1.1%)	45–64 (46.7%) 25–44 (26%) Others (27.3%)	45–64 (53.3%) 65–79 (21.2%) 25–44 (18.8%) 18–24 (3.4%) 80+ (3.3%)	45–64 (48%) 65–79 (24.9%) 25–44 (23.2%) 80+ (3%) 18–24 (0.6%)	25–44 (52.2%) 45–64 (34.8%) 65–79 (8.7%) 18–24 (4.3%)
Gender (M/F)	M - (57.9%) F - (42.1%)	M - (57.3%) F - (42.7%)	M - (51.2%) F - (48.6%)	M - (50.8%) F - (49.0%)	M - (56.5%) F - (43.5%)
Education	HS - (53.7%) 2-yr college - (27.4%) Others - (18.9%)	HS - (43.6%) 2-yr college - (22%) Others - (34.4%)	HS - (38.5%) 2-yr college - (37.7%) Others - (23.8%)	HS - (34.8%) 2-yr college - (29.7%) Uni - - (12.9%) Elem - (9.8%) Grad - (5.8%)	HS - (65.2%) 2-yr college - (26.1%) Others - (8.7)

Main Income Bracket	\$5,000-\$9,999 (23.2%) \$10K-\$19,999 (23.2%)	\$10K-\$19,999 (28.6%) \$20K-\$29,999 (16.3%)	\$10K-\$19,999 (21%) \$20K-\$29,999 (20.3%)	\$5,000-\$9,999 (18.7%) \$10K-\$19,999 (26.2%)	\$5,000-\$9,999 (17.4%) \$10K-\$19,999 (21.7%)
Main Income Source	Gov't work Gov't benefits	Gov't work Gov't benefits	Gov't work Gov't benefits	Gov't work Gov't benefits	Gov't work Other

Analysis of Regional Results Across Questionnaire Categories

Sustainable livelihood and food security

Households in the southwest region participated more in offshore pelagic fishing when compared to the East, South, and West regions, on average (Table D1). For those who sold reef fish, households in the West region were more likely to sell their fish in a fish market when compared to the South region, on average. For those who sold reef fish, households in the North region were more likely to sell their fish to the Northern Reef Fisheries Cooperative (NRFC) when compared to the East, South, and West regions, on average (Table D2). In the past 12 months, those in the West region were more likely to have used fish for customary and traditional events when compared to the East and South regions, on average (Table D3).

For those who eat offshore pelagic fish, households in the East region are more likely to prefer offshore pelagic fish because it provides more meat/feeds more people when compared to those in the South region, on average (Table D4).

For those who eat reef fish, households in the South region are more likely to prefer reef fish due to price when compared to those in the East region, on average. Households in the East and South regions are more likely to prefer reef fish because of availability when compared to those in the West region, on average. Additionally, households in the East region are more likely to prefer reef fish because it is high quality food when compared to those in the South region, on average. Households in the West region are more likely to prefer reef fish because it provides more meat/feed more people when compared to those in the South region, on average (Table D5).

In terms of where people source the pelagic fish that their household consumes, those in the North and Southwest regions are more likely to catch their own pelagic fish when compared to those in the East region, on average (Table D6). Those in the southwest region are also more likely to catch their own pelagic fish when compared to those in the South region, on average. Those in the West region are more likely to have bought imported pelagic fish from a fish market when compared to those in the South region, on average. Those in the South and East regions are more likely to have bought local fish from a supermarket when compared to those in the North region, on average. Those in the East and South regions are more likely to have bought their pelagic fish from Belau Offshore Fisheries Incorporated (BOFI) when compared to those in the

West region, on average. Those in the South region are also more likely to have bought their pelagic fish from BOFI when compared to those in the North region, on average. Those in the North region are more likely to have bought their pelagic fish from NRFC when compared to those in the West, East, and South regions, on average. Those in the South region are more likely to have bought their pelagic fish from a restaurant when compared to those in the West region, on average. Those in the South region are more likely to have received pelagic fish given to them by friends or neighbors when compared to those in the East region, on average.

In terms of where people source the reef fish that their household consumes, those in the North regions are more likely to catch their own reef fish when compared to those in the South region, on average (Table D7). Those in the west region are more likely to have bought imported reef fish from a fish market when compared to those in the North, South, and East regions, on average. Those in the east region are more likely to have bought imported reef fish from a fish market when compared to those in the south region, on average. Those in the West, South and East regions are more likely to have bought local fish from a supermarket when compared to those in the North region, on average. Those in the East region are more likely to have bought imported fish from a supermarket and bought from BOFI when compared to those in the South region, on average. Those in the North region are more likely to have bought their reef fish from NRFC when compared to those in the West, East, and South regions, on average.

With answers of “not sure” coded as missing, households in the Southwest region are more likely to consume fresh non-tuna pelagic offshore fish at least once a week when compared to those in the East, South, and West regions, on average. Those in the East, North, and South regions are more likely to consume fresh reef fish at least once a week when compared to those in the West region, on average. Those in the East region are more likely to consume other meat (e.g. pork, beef, chicken) at least once a week when compared to those in the North region, on average (Table D8).

Perceived benefits of the PNMS

With answers of “not sure” coded as missing the only statement where there was a difference in agreement across the region was “PNMS helps maintain Palau’s marine resources that are important to our sense of place and identity.” Households in the South agreed more with this statement than households in the East, on average (Table D9).

PNMS knowledge and support

Those in the West region are more likely to have no knowledge of the PNMS when compared to those in the East, North, and South regions, on average (Table 7). For those who have heard, seen, or read anything about the PNMS, those in the West region were more likely to find out through a community/town hall meeting when compared to those in the South region, on average (Table D10).

Table 7. Regional comparison of knowledge of the PNMS.

	Has any level of knowledge of PNMS (n=1,276)	Is at least moderately knowledgeable about PNMS (n=1,276)
East	63%	21%
North	66%	29%
South	68%	25%
Southwest	87%	26%
West	40%	16%
Test for difference (p<0.05)	E>W, N>W, S>W	

Discussion

This study presents the results of a Palau-wide socioeconomic survey that is intended to gain a better understanding of the human dimensions of the PNMS. The main objectives of this study are to 1) Understand the Palauan residents' thoughts about the PNMS, 2) understand how they do or do not believe they benefit from PNMS, 3) gather suggestions for improvement directly from the community, and 4) use this information to inform the management of the PNMS.

Nationwide analysis:

The general public perception of the PNMS is positive. Based on the survey, 70% of Palauan households supported the establishment of the PNMS and its objectives. However, the sentiment tends to be that it is a net good for certain population groups such as the greater community or future generations. A small group felt that the PNMS had negatively impacted them, and this was mainly from unintended consequences of the law, such as increased pelagic fish prices, loss of jobs in commercial fishery and fishery-adjacent sectors, inability to access protected areas, and inability to export fish. Although the respondents did not specify which type of fish cannot be exported, currently there is a law in place (RPPL10-54) that prohibits the export of reef fish but allows the export of pelagic fish. This law was put in place to alleviate the pressure on reef resources.

Notably, reef fish were preferred locally over pelagic fish due to its affordability, accessibility, and taste. Whether bought at the fish market, from local individuals, BOFI, or the state fisheries cooperatives, the average cost of pelagic fish reported was \$6.60 per pound. Approximately two-thirds of the respondents who purchase pelagic fish considered this price unaffordable. Moreover, buyers of pelagic fish also noted that it was not consistently available from their usual sellers. This indicates that pelagic fish are not only unaffordable, but they are also not accessible to the local community. A study by Lewis et al., 2020 found that the establishment of the PNMS may increase the price of offshore pelagic fish which may lead to a higher consumption of reef fish by tourists and incoming visitors. Therefore, to alleviate negative impacts on reef resources, efforts should be made to make pelagic fish more accessible to the community and the tourism industry.

There is a local distinction among reef fish, nearshore pelagic fish, offshore deepwater fish, and offshore pelagic fish, but the data from this study does not capture that. Survey findings from fishermen suggested that this distinction also shapes local understanding of the reef/pelagic binary. The literature similarly highlights a gendered understanding of nearshore fishing as the domain of men, yet when asked about reef fishing, several households mentioned doing “reef fishing” when they meant gleaning for sea cucumbers, clams, and other similar fauna, which are conducted primarily by women (Singeo et al., 2021). A more in-depth exploration of these cultural and social distinctions may lead to new insights into some patterns observed in this report.

Regional analysis:

This study indicated regional differences in pelagic fishing preferences and fish consumption. Households in the Southwest regions showed the highest participation in offshore pelagic fishing in comparison to the other regions. This may be due to the accessibility to the pelagic environment as well as the high abundance of pelagic fish. Goodell et al., 2024 indicated that the Southwest region had the highest abundance of Yellowfin tuna (*Thunnus albacares*) in comparison to other sites in Palau as well as around the world. Similar to the Southwest region, the East region also has easy access to pelagic fishing grounds, which may contribute to the observed preference of pelagic fish over reef fish in the East region households.

Moreover, the households in the North and Southwest regions were more likely to catch their own pelagic fish, indicating that local availability and fishing capability play an important role in shaping household consumption patterns. Moreover, the North region has a well-developed fisheries cooperative called the Northern Reef Fisheries Cooperative (NRFC) that supplies pelagic fish to the community. This survey indicates that households from all regions have previously purchased pelagic fish at NRFC.

Conclusion and Recommendations

In conclusion, results of this study revealed that a majority of the respondents support the PNMS viewing its implementation as a means to increase marine resources and safeguard the nation's supply of seafood. Moreover, this assessment indicated that there are existing misperceptions of PNMS and many people have little to no knowledge about it. Respondents also raised concerns about the increase in fish prices and inaccessibility of pelagic fish. Based on these findings, below are recommendations for improvement:

- To make pelagic fish more accessible to the community, Fish Aggregating Devices be strategically placed in each state to make pelagic fish more readily available and reduce pressure on reef fishes.
- Efforts should also focus on reducing the cost of pelagic fish to the local consumers as cost is a major barrier to food security.
- Enhanced community outreach should also be done to inform the community of what PNMS is, its boundaries, and its objectives as there were misperceptions about it.
- Follow-up studies should be done to assess the effectiveness of the recommendations above.

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Appendices

Appendix A. Tables of Sample Sizes

Table A1. Sample sizes and sampling weights for nationwide stratified sample of households of Palau.

State	Sample size	Total Household Population	Sample proportion	Population proportion	Sampling weight
Aimeliik	88	107	6.89%	2.14%	0.31
Airai	187	719	14.64%	14.37%	0.98
Angaur	47	49	3.68%	0.98%	0.27
Hatohobei	13	9	1.02%	0.18%	0.18
Kayangel	14	15	1.10%	0.30%	0.27
Koror	309	3162	24.20%	63.18%	2.61
Melekeok	52	94	4.07%	1.88%	0.46
Ngaraard	100	128	7.83%	2.56%	0.33
Ngarchelong	81	114	6.34%	2.28%	0.36
Ngardmau	32	72	2.51%	1.44%	0.57
Ngatpang	60	75	4.70%	1.50%	0.32
Ngchesar	71	101	5.56%	2.02%	0.36
Ngeremlengui	47	103	3.68%	2.06%	0.56
Ngiwal	57	89	4.46%	1.78%	0.40
Peleliu	109	154	8.54%	3.08%	0.36
Sonsorol	10	14	0.78%	0.28%	0.36
Total	1,277	5,005	100%	100%	

Table A2. Sample sizes and sampling weights for regions of Palau (95% confidence level, 5% confidence interval).

Region	States	Sample size	Total household population	Sample proportion	Population proportion	Sampling weight by region
North	Kayangel Narchelong	95	129	7.44%	2.58%	0.35
East	Ngaraard Ngiwal Melekeok	467	1131	36.57%	22.60%	0.62

	Ngchesar Airai					
West	Ngardmau Ngeremlengui Ngatpang Aimeliik	227	357	17.78%	7.13%	0.40
South	Koror Peleliu Angaur	465	3365	36.41%	67.23%	1.85
Southwest	Sonsorol Hatohobei	23	23	1.80%	0.46%	0.26
TOTAL		1,277	5,005	100.00%	100.00%	1.00

Appendix B. Nationwide results (Tables and Figures)

Socio-demographics for nationwide sample

Table B1. Education level of respondents nationwide, calculated with sampling weights (n=1277).

Level of Education	Weighted Mean	Number of Respondents
Elementary	5.0%	87
High School	35.1%	507
Vocational School/Training	6.0%	102
Some College	32.7%	368
University	12.8%	133
Master's or PhD	6.3%	50
None	0.5%	13
Refused to answer	1.5%	17

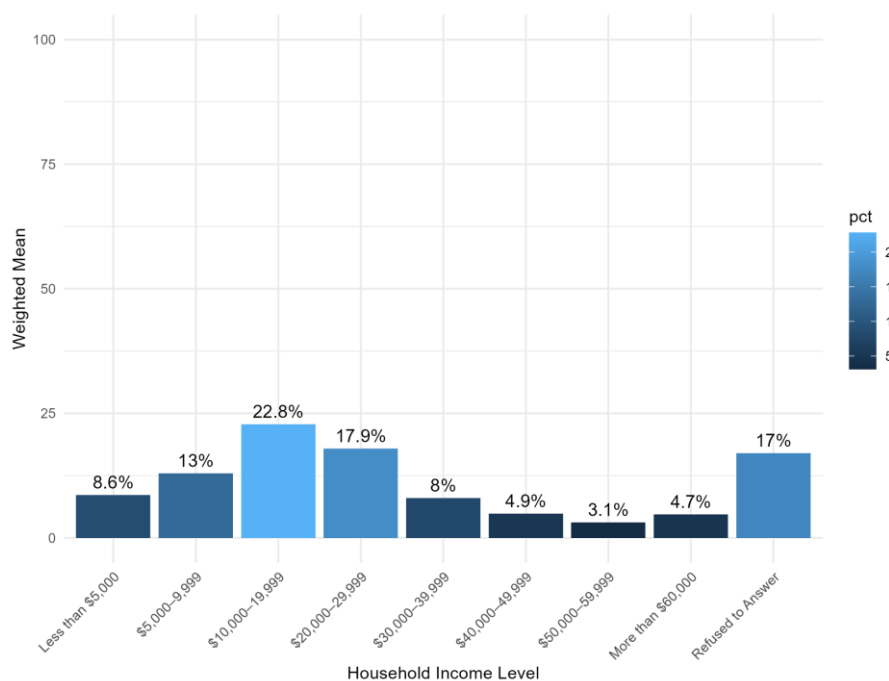


Figure B1. Reported nationwide household income level, with stratification weights (n=1277). Standard error for all measures was very small thus not displayed on plot ($\leq 0.01\%$).

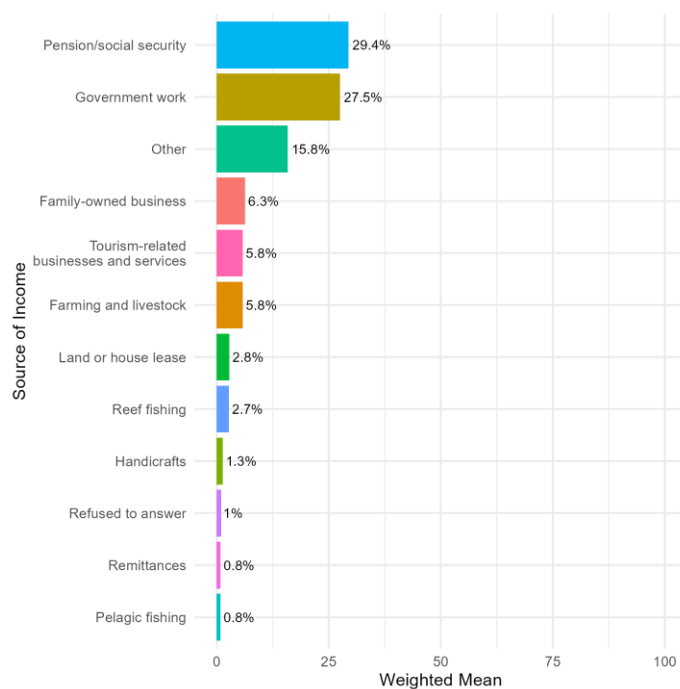


Figure B2. Top three sources of income reported nationwide, by weighted mean (n=1277).

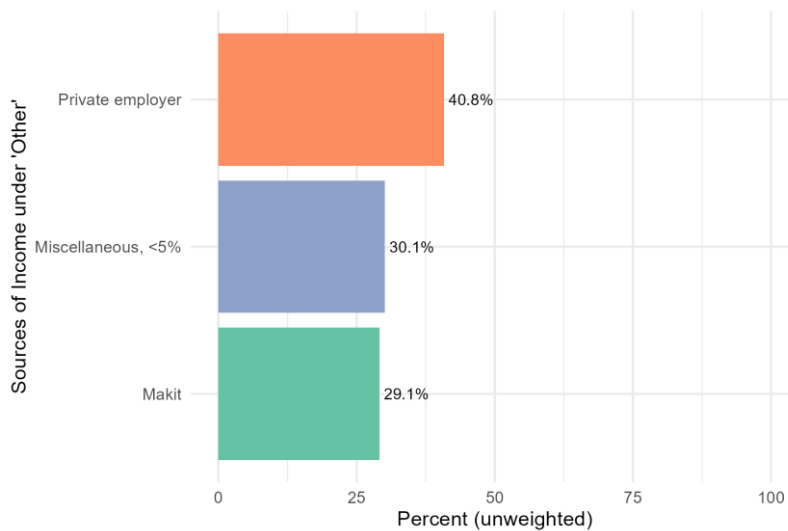


Figure B3. Sources of income listed under “Other” reported by residents of Palau nationwide (n=297).

PNMS knowledge and support

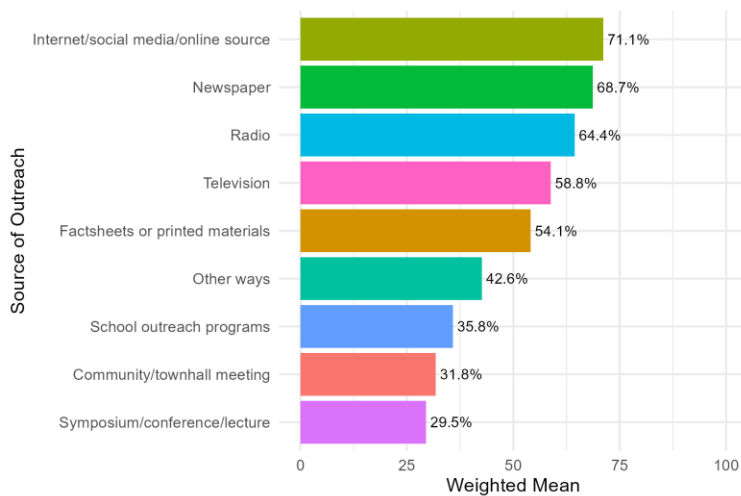


Figure B4. Source of outreach regarding the PNMS for respondents nationwide. Standard error is <0.1% for all.

Perceived benefits of the PNMS

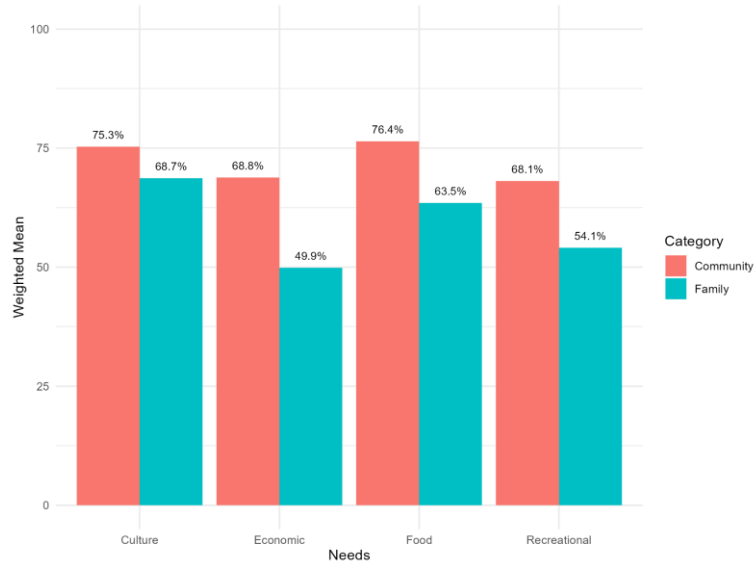


Figure B5. Answer to the question, “Were your family and community needs taken into the planning and consideration of the PNMS?” among respondents nationwide. Standard error is <0.1% for all.

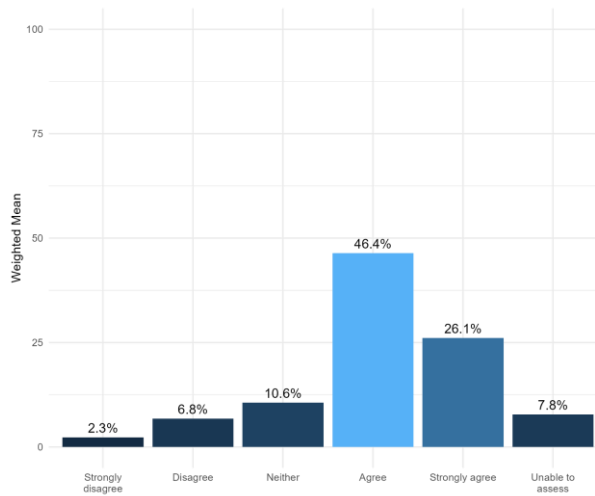


Figure B6. Percent of respondents nationwide who believe that the PNMS benefits all people of Palau equitably. Standard error is <0.1% for all.

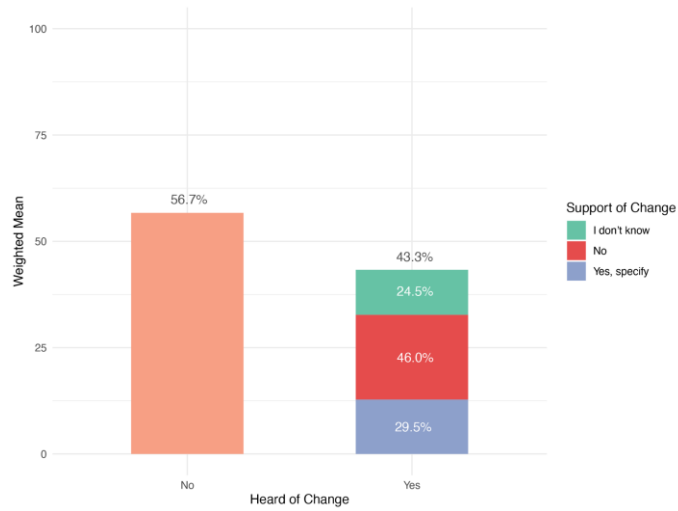


Figure B7. Nationwide awareness of changing the PNMS law. Those who answered “yes” were then asked if they support the change, as displayed in the “yes” bar of the chart. Standard error for all weighted responses was <0.1%.

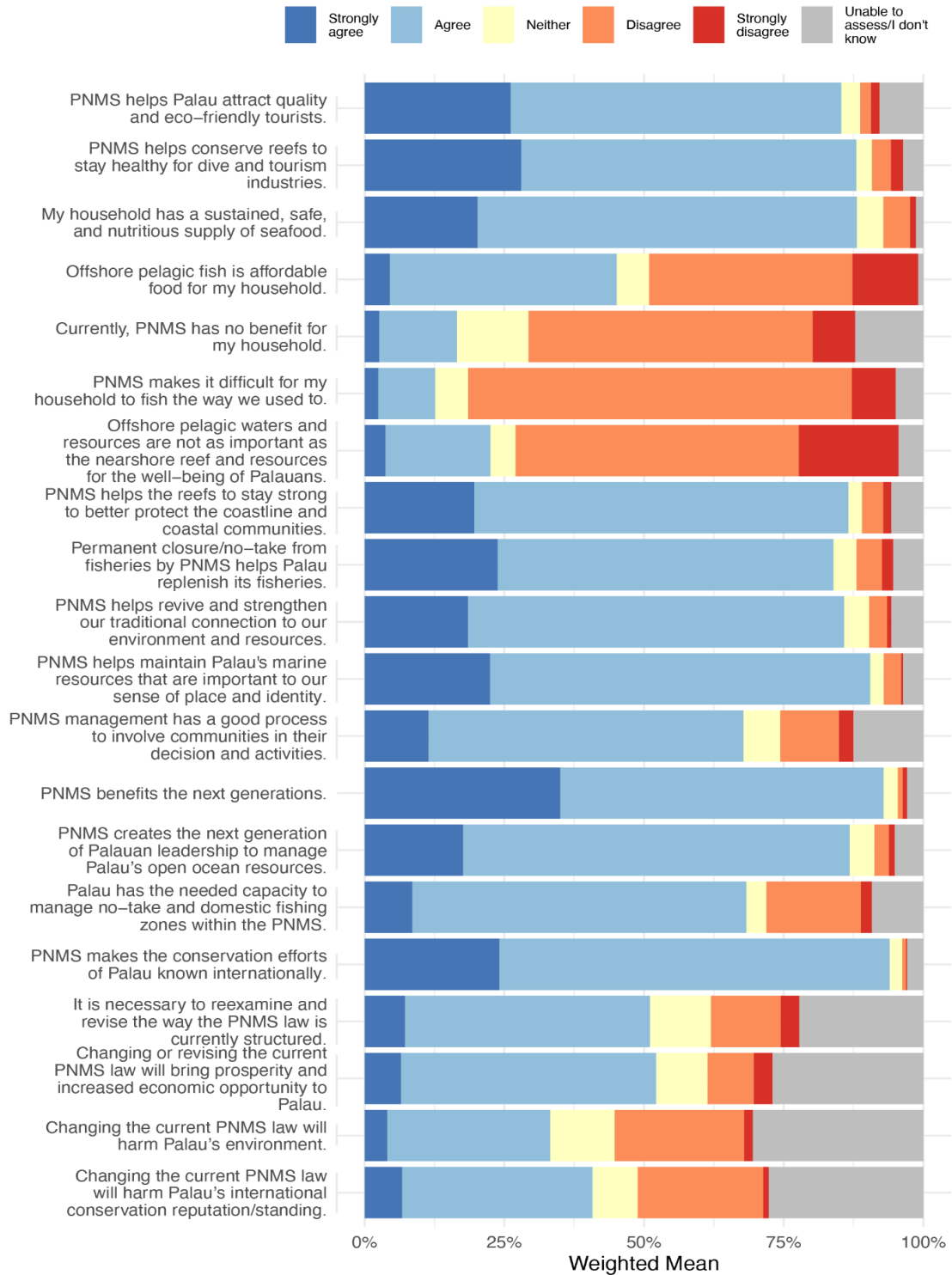


Figure B8. Level of agreement nationwide to statements regarding the PNMS, weighted by state.

Sustainable livelihood and food security

Table B2. Businesses, establishments, and people that buy pelagic fish, and the average price they pay, based on reports from pelagic fishers nationwide who sell their catch.

Buyer (Pelagic)	Weighted %	SE ($\pm$ %)	n (Buyer)	Average Price (\$)	Price SE ($\pm$ \$)	n (Price)
Community/individuals	45.6	± 0.1	42	2.80	± 0.22	41
Fish Market	35.4	± 0.1	28	4.40	± 0.88	26
Restaurant	31.6	± 0.1	24	5.30	± 1.12	21
Makit	30.6	± 0.1	37	4.70	± 1.12	30
Hotel Restaurant	26.0	± 0.1	15	3.30	± 0.23	13
Supermarket	14.7	$\pm < 0.1$	18	5.50	± 1.74	16
NRFC	13.7	$\pm < 0.1$	16	3.40	± 0.08	15
BOFI	12.6	± 0.1	7	2.80	± 0.18	7
Caterers	12.4	$\pm < 0.1$	13	3.60	± 0.6	12
Others	6.5	$\pm < 0.1$	6	3.10	± 0.7	6
Government	3.7	$\pm < 0.1$	6	3.50	± 1.05	3

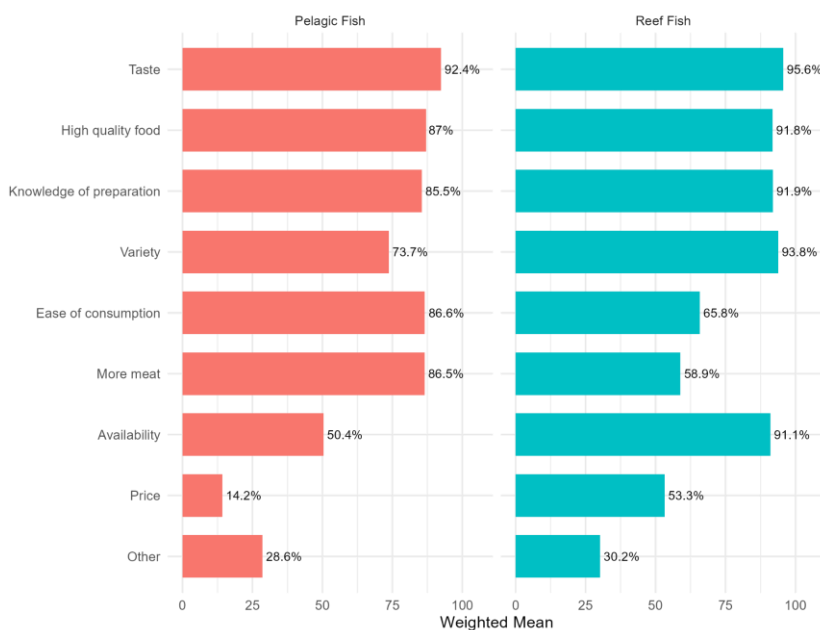


Figure B9. Reasons for preference of type of fish, as reported nationwide.

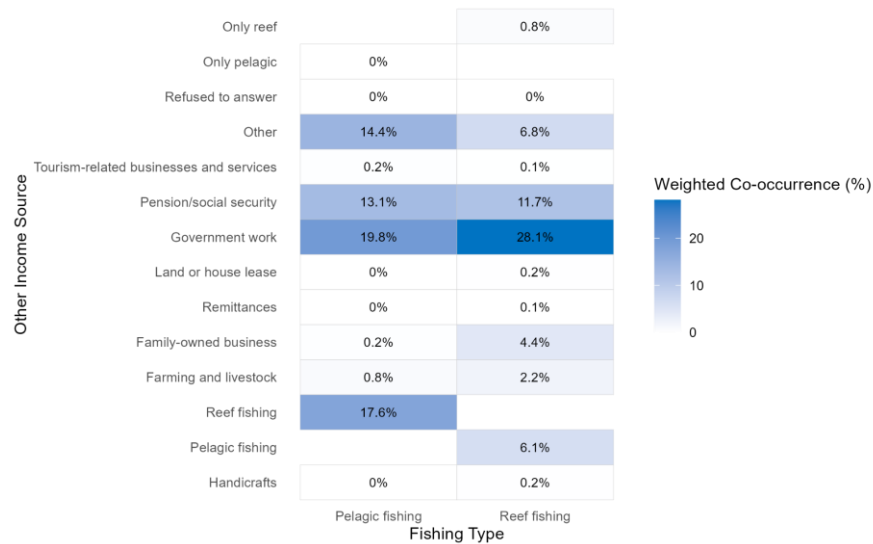


Figure B10. Weighted percent of respondents nationwide who selected pelagic or reef fishing as a top three source of income, co-occurring with another source of income.

Appendix C. Analysis of Nationwide Results Across Questionnaire Categories Tables

Table C1. Cross analysis of household knowledge of the PNMS with participation in pelagic offshore fishing, with post-stratification weights at the state level applied.

	Household does not participate in pelagic offshore fishing		Household participates in pelagic offshore fishing		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Has any level of knowledge of PNMS	1,030	64%	244	86%	-7.95	<0.01
Is at least moderately knowledgeable about PNMS	1,030	21%	244	41%	-5.97	<0.01
Has heard, seen, or read anything about PNMS, and/or participated in any outreach or education activities related to the PNMS	666	61%	209	76%	-4.12	<0.01

Table C2. Cross analysis of household knowledge of the objectives of the PNMS, with participation in pelagic offshore fishing. Post-stratification weights at the state level are applied.

Answered correctly about the following objectives	Household does not participate in pelagic offshore fishing		Household participates in pelagic offshore fishing		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Healthy ocean	402	52%	159	52%	-0.12	0.91
Healthy coral reef	402	33%	159	35%	-0.51	0.61
Benefit Palau's economy	402	25%	159	34%	-2.03	0.04
Benefit Palauan's livelihood	402	32%	159	44%	-2.59	<0.01
Increase marine resources	402	58%	159	58%	0.11	0.91
Maintain supply of nutritious and safe fish (seafood) for everyone	402	51%	159	60%	-1.86	0.06
Develop domestic pelagic fishing industry	402	22%	159	33%	-2.67	<0.01
Support existing sustainable tourism	402	20%	159	24%	-1.02	0.31
Sustainable Development	402	42%	159	54%	-2.78	<0.01

Table C3. Cross analysis of two factors: household perception of inclusion in the implementation of the PNMS, and participation in pelagic offshore fishing. Post stratification weights at the state level applied.

Based on the list of needs below, please indicate whether you feel your family and community needs were taken into consideration during the planning and implementation of the PNMS?	Household does not participate in pelagic offshore fishing		Household participates in pelagic offshore fishing		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Family Food needs	294	78%	137	88%	-2.86	<0.01
Family Economic needs	274	63%	124	82%	-4.36	<0.01
Family Recreation needs	294	63%	132	85%	-5.18	<0.01
Family Cultural needs	318	81%	134	89%	-2.25	0.03
Community Food needs	332	87%	140	93%	-2.23	0.03
Community Economic needs	296	83%	139	94%	-3.81	<0.01
Community Recreation needs	292	84%	136	94%	-3.37	<0.01
Community Cultural needs	317	90%	138	93%	-1.07	0.29

Table C4. Cross analysis of households who reported impacts from COVID-19 with participation in pelagic offshore fishing. Post stratification weights by state applied.

	Household does not participate in pelagic offshore fishing		Household participates in pelagic offshore fishing		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
In the past 12 months, have impacts from COVID-19 altered your fishing behavior?	1,018	17%	243	31%	-4.31	<0.01

Table C5. Cross analysis of households with households which are aware of, received outreach about, and support the PNMS, with participation in reef fishing. Post stratification weights by state applied.

	Household does not participate in reef fishing		Household participates in reef fishing		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Has any level of knowledge of PNMS	559	58%	715	77%	-7.37	<0.01
Is at least moderately knowledgeable about PNMS	559	19%	715	29%	-4.30	<0.01
Has heard, seen, or read anything about PNMS, and/or participated in any outreach or education activities related to the PNMS	325	59%	550	68%	-2.83	<0.01
Supports or strongly supports the establishment of the 80% of Palau's Exclusive Economic Zone (EEZ) as a no-take zone	176	69%	348	79%	-2.50	0.01

Table C6. Cross analysis of two factors: household perception of inclusion in the implementation of the PNMS, and participation in reef fishing. Post stratification weights at the state level applied. For analysis, answers of “not sure” were coded as missing.

Based on the list of needs below, please indicate whether you feel your family and community needs were taken into consideration during the planning and implementation of the PNMS?	Household does not participate in reef fishing		Household participates in reef fishing		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Family Food needs	138	77%	292	83%	-1.22	0.22
Family Economic needs	118	64%	280	71%	-1.27	0.21
Family Recreation needs	117	62%	309	73%	-2.15	0.03
Family Cultural needs	136	77%	316	86%	-2.17	0.03
Community Food needs	159	88%	314	89%	-0.43	0.67
Community Economic needs	147	85%	288	88%	-0.76	0.45

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Community Recreation needs	144	88%	284	87%	0.09	0.93
Community Cultural needs	151	93%	304	90%	1.1	0.27

Table C7. Cross analysis of household agreement to statements regarding the PNMS and participation in reef fishing. Post stratification weights by state applied. During analysis, answers of “not sure” were coded as missing, and results that are not statistically significant are not reported.

Please rate your agreement with the following statements	Household does not participate in reef fishing		Household participates in reef fishing		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
My household has a sustained, safe, and nutritious supply of seafood.	187	3.94	364	4.07	-2.04	0.04
PNMS helps the reefs to stay strong to better protect the coastline and coastal communities	178	4.15	356	4.01	2.04	0.04
Permanent closure or no-take from fisheries by PNMS helps Palau sustain and replenish its fisheries	177	4.20	359	3.97	3.03	<0.01
Changing the current PNMS law will harm Palau's international conservation reputation/standing	132	3.56	274	3.20	3.26	<0.01

Table C8. Cross analysis of households who reported impacts from COVID-19 with participation in reef fishing. Post stratification weights by state applied.

	Household does not participate in reef fishing		Household participates in reef fishing		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
In the past 12 months, have impacts from COVID-19 altered your fishing behavior?	553	11%	708	27%	-7.79	<0.01

Table C9. Cross analysis of households who reported having any knowledge of PNMS or had received outreach about PNMS, and use of fish in customary or traditional events. Post stratification weights by state applied.

	Household does not use fish in customary or traditional events		Household uses fish in customary or traditional events		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Has any level of knowledge of PNMS	534	56%	737	78%	-8.33	<0.01
Is at least moderately knowledgeable about PNMS	534	14%	737	32%	-7.76	<0.01
Has heard, seen, or read anything about PNMS, and/or participated in any outreach or education activities related to the PNMS	298	52%	575	72%	-5.82	<0.01

Table C10. Cross analysis across channels of outreach regarding the PNMS and household use of fish in traditional or customary events. Post stratification weights by state applied.

Has heard, seen, or read anything about PNMS, and/or participated in any outreach or education activities related to the PNMS from:	Household does not use fish in customary or traditional events		Household uses fish in customary or traditional events		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Factsheet	154	46%	411	57%	-2.29	0.02
School outreach programs (Earth Day fairs, PICRC)	154	35%	411	36%	-0.37	0.71
Television (TV)	154	45%	412	64%	-4.16	<0.01
Newspaper	154	61%	411	72%	-2.37	0.02
Radio	154	46%	411	71%	-5.50	<0.01
Community/town hall meeting	154	22%	411	36%	-3.44	<0.01
Symposiums/conferences/lectures	154	25%	411	31%	-1.48	0.14
Internet/Social media or online sources (Facebook, Instagram, Google)	154	65%	412	73%	-1.89	0.06

Table C11. Cross analysis of household awareness of objectives of the PNMS and use of fish in customary or traditional events. Post stratification weights by state applied.

Answered correctly about the following objectives	Household does not use fish in customary or traditional events		Household uses fish in customary or traditional events		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Healthy ocean	149	51%	411	52%	-0.34	0.73
Healthy coral reef	149	31%	411	34%	-0.59	0.55
Benefit Palau's economy	149	22%	411	30%	-1.79	0.08
Benefit Palauan's livelihood	149	28%	411	38%	-2.28	0.02
Increase marine resources	149	48%	411	62%	-2.91	<0.01
Maintain supply of nutritious and safe fish (seafood) for everyone	149	42%	411	58%	-3.34	<0.01
Develop domestic pelagic fishing industry	149	15%	411	28%	-3.39	<0.01
Support existing sustainable tourism	149	19%	411	22%	-0.65	0.52
Sustainable Development	149	37%	411	48%	-2.35	0.02

Table C12. Cross analysis of agreement with statements regarding the PNMS and use of fish in customary or traditional events. Post stratification weights by state applied. During analysis, answers of “not sure” were coded as missing, and results that are not statistically significant are not reported.

Please rate your agreement with the following statements	Household does not use fish in customary or traditional events		Household uses fish in customary or traditional events		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
My household has a sustained, safe, and nutritious supply of seafood.	139	3.88	379	4.08	-2.82	<0.01
PNMS makes it difficult for my household to fish the way we used to	135	2.47	365	2.20	2.94	<0.01
PNMS helps revive and strengthen our traditional connection to our environment and resources	141	3.96	391	4.09	-2.04	0.04
PNMS creates the next generation of Palauan leadership to manage Palau's open ocean resources.	146	3.92	392	4.10	-2.80	<0.01
Palau has the needed capacity to manage no-take and domestic fishing zones within the PNMS.	136	3.76	377	3.56	2.13	0.03
PNMS makes the conservation efforts of Palau known internationally.	151	4.09	396	4.25	-3.44	<0.01
It is necessary to reexamine and revise the way the PNMS law is currently structured.	121	3.72	317	3.42	3.13	<0.01

Table C13. Cross analysis of household knowledge of the PNMS with fishing activity. Post stratification weights by state applied.

	Household does not fish once a week or more		Household does fish once a week or more		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Has any level of knowledge of PNMS	663	62%	610	76%	-5.26	<0.01

Table C14. Cross analysis of agreement with statements regarding the PNMS with household fishing activity. Post stratification weights by state applied. Answers of “not sure” are coded as missing in the table, and results that are not statistically significant are not reported.

Please rate your agreement with the following statements	Household does not fish once a week or more		Household does fish once a week or more		Test for difference	
	Weighted n	Mean	Weighted n	Mean	t-statistic	p-value
Offshore pelagic fish is affordable food for my household.	259	3.02	283	2.78	2.29	0.02
Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans.	260	2.23	278	2.50	-2.85	<0.01

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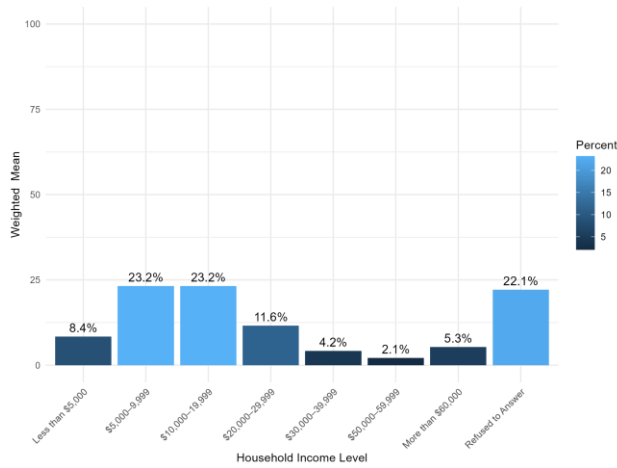
Permanent closure or no-take from fisheries by PNMS helps Palau sustain and replenish its fisheries	252	4.18	284	3.93	3.58	<0.01
It is necessary to reexamine and revise the way the PNMS law is currently structured.	190	3.37	248	3.60	-2.39	0.02
Changing or revising the current PNMS law will bring prosperity and increased economic opportunity to Palau.	189	3.48	222	3.70	-2.38	0.02
Changing the current PNMS law will harm Palau's environment.	182	3.36	209	2.98	3.69	<0.01

Table C15. Spearman correlation analysis of agreement with statements about the PNMS with reported channel of outreach regarding the PNMS. ** = statistically significant with 99% confidence; * = statistically significant with 95% confidence

	Fact sheets/ printed materials	School outreach programs	Television	Newspaper	Radio	Community meeting	Symposia	Internet/ Social media
PNMS helps Palau attract quality and eco-friendly tourists.						-0.10*		
PNMS helps conserve reefs to stay healthy for dive and tourism industries.						-0.12**		0.12**
My household has a sustained, safe, and nutritious supply of seafood.				0.09*			0.13**	
Offshore pelagic fish is affordable food for my household.								
Currently PNMS has no benefit for my household.							-0.13**	
PNMS makes it difficult for my household to fish the way we used to.								
Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans.			0.14**				-0.10*	-0.16**
PNMS helps the reefs to stay strong to better protect the coastline and coastal communities.						-0.13**		0.09*
Permanent closure or no-take from fisheries by PNMS helps Palau sustain and replenish its fisheries.	0.09*		-0.09*				0.09*	0.11*
PNMS helps revive and strengthen our traditional connection to our environment and resources.				0.09*			0.09*	
PNMS helps maintain Palau's marine resources that are important to our sense of place and identity.		0.09*		0.12**			0.09*	0.14**
PNMS management has a good process to involve communities in their decisions and activities.		0.09*	0.17**	0.13**	0.21**			
PNMS benefits the next generations.							0.18**	0.14**
PNMS creates the next generation of Palauan leadership to manage Palau's open ocean resources.	0.09*	0.11**	0.15*	0.16**	0.12**		0.10*	
Palau has the needed capacity to manage no-take and domestic fishing zones within the PNMS.			0.11*		0.11*		-0.10*	
PNMS makes the conservation efforts of Palau known internationally.	0.13**	0.11**		0.09*	0.11**		0.21**	0.12**
It is necessary to reexamine and revise the way the PNMS law is currently structured.								
Changing or revising the current PNMS law will bring prosperity and increased economic opportunity to Palau.		-0.12*			0.10*			-0.14**
Changing the current PNMS law will harm Palau's environment.								
Changing the current PNMS law will harm Palau's international conservation reputation/standing.				0.12*				

Appendix D. Regional Results

a.



b.

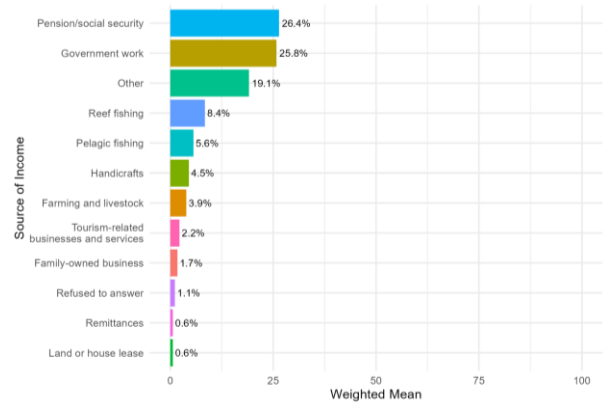
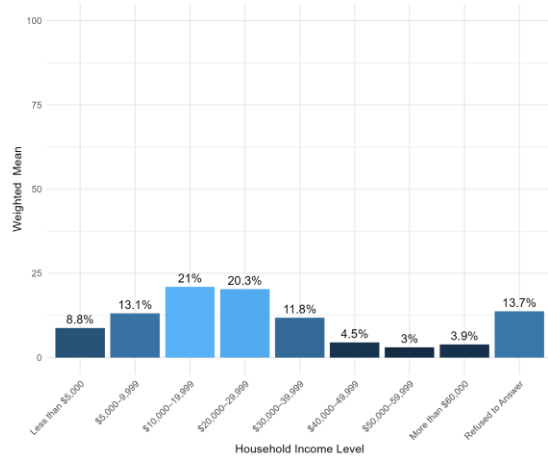


Figure D1a. Distribution of income levels across the northern region of Palau, D1b. Top three sources of income in households in the northern region.

a.



b.

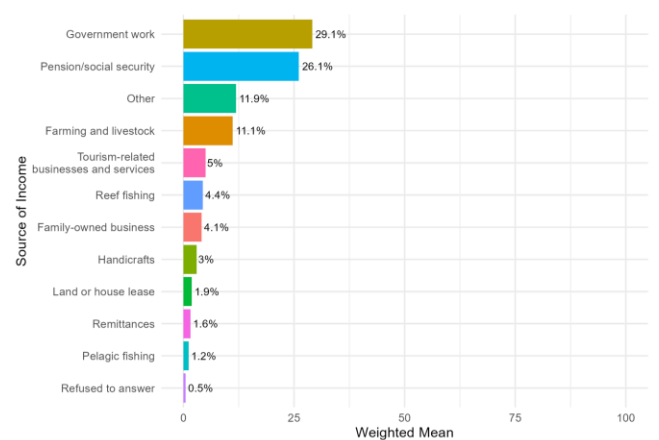
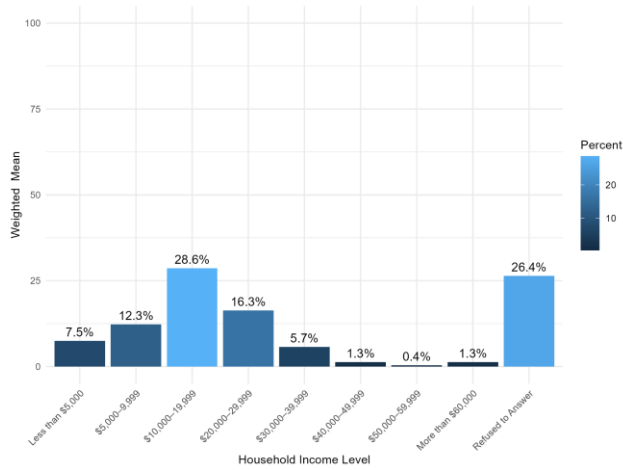


Figure 2Da. Distribution of income levels across the eastern region of Palau, D2b. Top three sources of income in households in the eastern region.

a.



b.

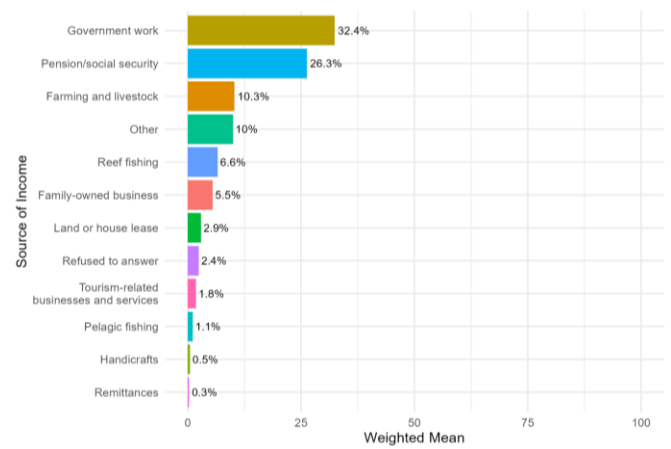
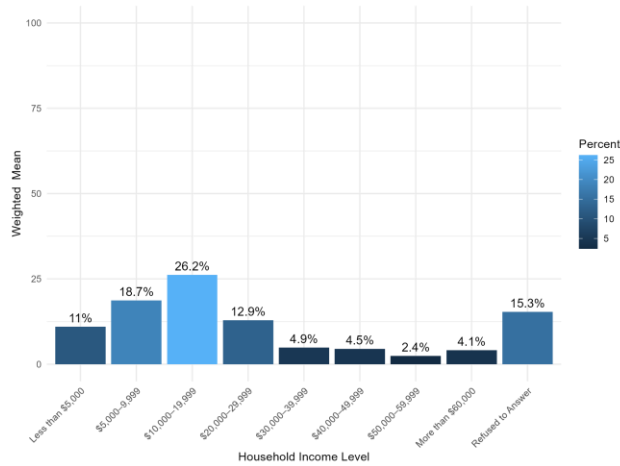


Figure D3a. Distribution of income levels across the western region of Palau, D3b. Top three sources of income in households in the western region.

a.



b.

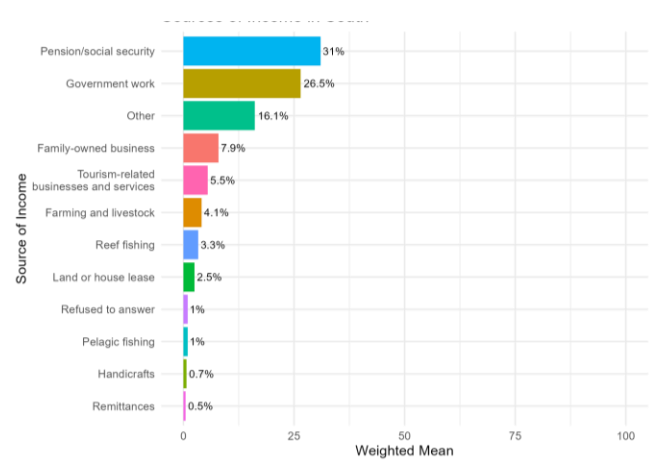
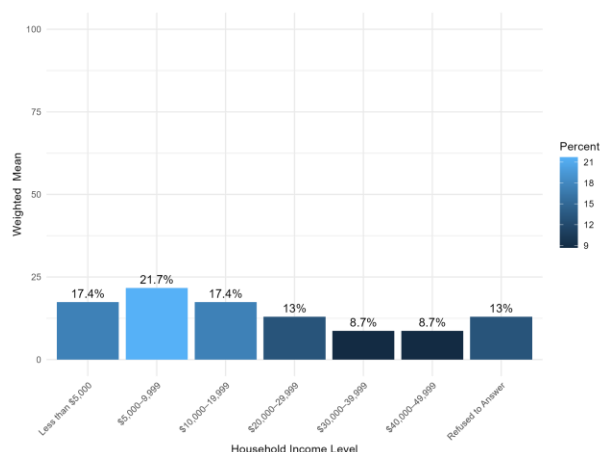


Figure D4a. Distribution of income levels across the southern region of Palau, D4b. Top three sources of income in households in the southern region.

a.



b.

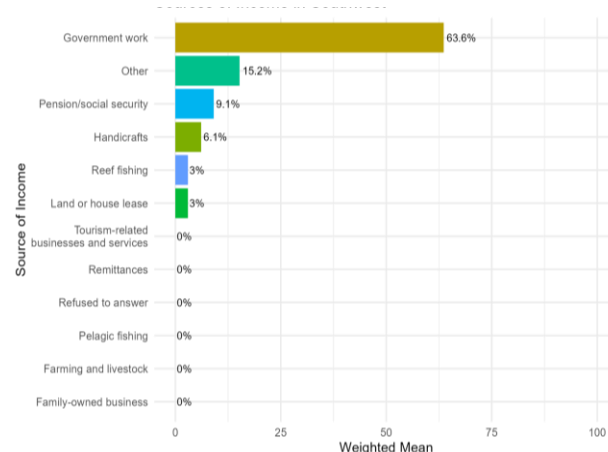


Figure D5a. Distribution of income levels across the southwestern region of Palau, D5b. Top three sources of income in households in the southwestern region.

Table D1. Regional comparison of household participation in various types of fishing and frequency. Post stratification weights by region applied.

Region	Household participates in offshore pelagic fishing (n=1,275)	Household participates in reef fishing (n=1,275)	Household participates in any kind of fishing (n=1,275)	Household participates in any kind of fishing once a week or more (n=1,275)
East	18%	59%	60%	50%
North	34%	73%	74%	63%
South	21%	56%	59%	48%
Southwest	83%	96%	96%	83%
West	26%	63%	63%	53%
Test for difference (p<0.05)	SW>E, S and W			

Table D2. Regional comparison of reef fish buyers, with post stratification weight by region applied.

For those who sell reef fish, who do you sell it to?	East	North	South	Southwest	West	Test for difference (p<0.05)
Fish market (n=163)	25%	22%	28%	100%	66%	W>S
Supermarket (n=162)	20%	6%	21%	100%	35%	
Hotel Restaurant (n=162)	8%	6%	16%	0%	18%	
Restaurant (n=162)	15%	11%	14%	0%	20%	
Directly to community/individuals (n=162)	70%	56%	72%	100%	33%	

Belau Offshore Fisheries Incorporated (BOFI) (n=162)	0%	0%	2%	0%	0%	
Northern Reef Fisheries Cooperative (NRFC) (n=162)	2%	61%	0%	0%	5%	N>E, S, and W
Government (MOE, MOJ, MHHS, etc) (n=162)	3%	6%	7%	0%	2%	
Caterers (n=162)	15%	17%	30%	0%	8%	
Makit (n=163)	52%	22%	35%	100%	68%	

Table D3. Regional comparison of household use of fish for customary or traditional events, with post stratification weights by region applied.

Region	In the past 12 months, how often has pelagic offshore and/or reef fish been used by your household for customary and traditional events? (n=1,273)
East	59%
North	68%
South	56%
Southwest	57%
West	76%
Test for difference (p<0.05)	W>E and S

Table D4. Regional comparison of reasons for preference of pelagic fish, with post stratification weights by region applied.

Please explain why your household prefers eating pelagic offshore fish	East	North	South	Southwest	West	Test for difference (p<0.05)
Price (n=668)	12%	23%	14%	28%	16%	
Availability (n=668)	47%	52%	50%	83%	39%	
Taste preference (n=668)	91%	88%	94%	89%	96%	
High quality food (n=668)	90%	90%	88%	89%	95%	
Knowledge of how to prepare (n=667)	89%	83%	86%	94%	95%	
Variety (n=668)	79%	81%	72%	78%	79%	
Ease of consumption (n=668)	93%	92%	84%	89%	92%	
More meat/feeds more people (n=668)	93%	83%	82%	94%	98%	E>S

Table D5. Regional comparison of reasons for preference of reef fish, with post stratification weights by region applied.

Please explain why your household prefers eating reef fish	East	North	South	Southwest	West	Test for difference (p<0.05)
Price (n=1,158)	46%	43%	56%	48%	47%	S>E
Availability (n=1,158)	90%	90%	92%	90%	79%	E>W; S>W
Taste preference (n=1,158)	96%	100%	94%	100%	99%	
High quality food (n=1,158)	97%	98%	88%	100%	96%	E>S
Knowledge of how to prepare (n=1,157)	95%	98%	90%	100%	98%	
Variety (n=1,158)	96%	97%	92%	100%	98%	
Ease of consumption (n=1,158)	68%	74%	65%	71%	70%	
More meat/feeds more people (n=1,157)	62%	57%	54%	62%	72%	W>S

Table D6. Regional comparison of pelagic offshore fish source for households, with post stratification weights by region applied.

Where do you get fish for food in your household? - Pelagic offshore fish	East	North	South	Southwest	West	Test for difference (p<0.05)
Caught by myself or someone else in our household (n=1,146)	26%	51%	31%	87%	36%	N>E, SW>E, and S
Bought from fish market (local) (n=1,145)	36%	26%	46%	26%	38%	
Bought from fish market (imported) (n=1,145)	12%	9%	9%	0%	23%	W>S
Bought from supermarket (local) (n=1,145)	43%	15%	43%	17%	39%	S>N, E>N
Bought from supermarket (imported) (n=1,145)	17%	10%	16%	0%	25%	
Bought from Belau Offshore Fisheries Incorporated (BOFI) (n=1,145)	40%	16%	42%	30%	22%	E>W, S>N and W
Bought from Northern Reef Fisheries Cooperative (NRFC) (n=1,145)	20%	60%	16%	17%	14%	N>E, S, and W
Bought from a neighbor, friend, or other person (not from market or store) (n=1,146)	37%	24%	40%	13%	31%	
Restaurant (n=1,145)	35%	23%	41%	13%	23%	S>W
Given by family or friends (n=1,145)	81%	80%	88%	91%	79%	S>E

Table D7. Regional comparison of reef fish source for households, with post stratification weights by region applied.

Where do you get fish for food in your household? - Reef fish	East	North	South	Southwest	West	Test for difference (p<0.05)
Caught by myself or someone else in our household (n=1,216)	66%	84%	59%	96%	69%	N>S
Bought from fish market (local) (n=1,214)	44%	27%	50%	35%	54%	
Bought from fish market (imported) (n=1,214)	15%	5%	8%	0%	28%	E>S, W>N, S, and E
Bought from supermarket (local) (n=1,214)	44%	11%	39%	13%	44%	E>N, S>N, W>N
Bought from supermarket (imported) (n=1,214)	33%	21%	22%	4%	29%	E>S

Bought from Belau Offshore Fisheries Incorporated (BOFI) (n=1,214)	16%	4%	8%	9%	12%	E>S
Bought from Northern Reef Fisheries Cooperative (NRFC) (n=1,213)	16%	34%	11%	9%	9%	N>E, S and W
Bought from a neighbor, friend, or other person (not from market or store) (n=1,216)	44%	32%	48%	22%	44%	
Restaurant (n=1,214)	27%	18%	26%	9%	24%	
Given by family or friends (n=1,215)	91%	85%	92%	91%	84%	

Table D8. Regional comparison of frequency of household protein consumption, with post stratification weights by region applied.

On average in the past 12 months, does your household eat the following protein at least once a week?	East	North	South	Southwest	West	Test for difference (p<0.05)
Fresh Tuna (n=1,258)	5%	8%	8%	30%	5%	
Processed tuna (e.g. smoked, flakes, dried) (n=1,261)	3%	1%	4%	9%	4%	
Fresh non-tuna pelagic offshore fish (e.g. mahi mahi, wahoo, marlin) (n=1,247)	3%	7%	5%	30%	3%	SW>E, S and W
Processed non-tuna (e.g. smoked, flakes, dried) (n=1,250)	1%	2%	2%	13%	3%	
Imported fish (salmon, basa, tilapia) (n=1,255)	5%	5%	5%	0%	5%	
Fresh reef fish (n=1,259)	69%	81%	69%	74%	41%	E>W; N>W; S>W
Processed reef fish (e.g. smoked, flakes, dried) (n=1,256)	16%	24%	19%	30%	21%	
Canned meat (canned tuna, SPAM etc) (n=1,265)	55%	50%	63%	52%	67%	
Other protein/meat (i.e. pork, beef, chicken) (n=1,268)	82%	59%	78%	52%	79%	E>N

Table D9. Regional comparison of agreement to the statement, “PNMS helps maintain Palau’s marine resources that are important to our sense of place and identity.”

	PNMS helps maintain Palau’s marine resources that are important to our sense of place and identity (n=478)
East	3.99
North	3.76
South	4.19
Southwest	4.00
West	3.92
Test for difference (p<0.05)	S>E

Table D10. Regional comparison of methods of PNMS-related outreach.

For those who have heard, seen, or read anything about PNMS, and/or participated in any outreach or education activities related to the PNMS, from where did you hear about it?	East	North	South	Southwest	West	Test for difference (p<0.05)
Fact sheets/printed materials (brochures, posters, flyers) (n=494)	54%	48%	56%	58%	59%	
School outreach programs (Earth Day fairs, PICRC) (n=493)	40%	43%	33%	33%	52%	
Television (TV) (n=495)	64%	61%	55%	33%	75%	
Newspaper (n=494)	58%	52%	66%	67%	71%	
Radio (n=494)	63%	59%	60%	50%	69%	
Community/town hall meeting (n=494)	42%	59%	33%	25%	65%	W>S
Symposiums/conferences/lectures (n=493)	35%	34%	28%	8%	42%	
Internet/Social media or online sources (Facebook, Instagram, Google) (n=495)	64%	70%	67%	58%	81%	

Appendix E: Survey Questionnaire

University of Hawai'i

Consent to Participate in a Research Project

Project title: Understanding human benefits and values of the Palau National Marine Sanctuary to the residents of Palau

The Palau International Coral Reef Center (PICRC) kindly seeks your assistance with a research project to understand the expected values and benefits of the Palau National Marine Sanctuary (PNMS) to the residents of Palau. The results of the research will be communicated back to the communities as well as in reports. The information will be used to improve the management of the PNMS.

Activities and Time Commitment

The research involves collecting information using a survey questionnaire. The survey will take approximately 30 minutes to complete.

Benefits and Risks

The results will help inform how to improve the PNMS management, to maintain the natural resources of Palau and to benefit the livelihoods of the residents of Palau. The investigators believe there is minimal or no risk to participate in this research project.

Privacy and Confidentiality

None of your personal information will be included in the survey results and your input will be aggregated to the community level. The data we collect today will be stored securely and kept confidential. Only the University of Hawai'i principal investigator, the PICRC Research Department and the IT consultant will have access to the data. Other agencies that have legal permission have the right to review research records. The University of Hawai'i Human Studies Program has the right to review research records for this study. The completed questionnaires will be destroyed after the data analysis is completed.

Voluntary Participation

Your participation in the project is completely voluntary. You do not have to answer any question that you don't want to, and you may withdraw your participation at any time without consequences.

☐ Please check this box if you volunteer to participate in this survey.

Do you consent to audio taping during the interview? (Please check box)

☐ Yes

☐ No

Do you consent to photographs during the survey or interview? (Please check box)

☐ Yes

☐ No

Interviewee signature: _____

Date: _____

Signature of enumerator: _____

Date: _____

Keep this copy of the informed consent for your records and reference.

Questions

If you have any further questions about this study after the survey, please contact

Elsei D. Tellei at PICRC via phone 488-6950 or via email: elseitellei@gmail.com.

You may contact the University of Hawaii Human Studies Program at 808.956.5007 or uhirb@hawaii.edu. to discuss problems, concerns and questions; obtain information; or offer input with an informed individual who is unaffiliated with the specific research protocol.

Please visit <https://www.hawaii.edu/researchcompliance/information-research-participants> for more information on your rights as a research participant.

Please visit <https://www.hawaii.edu/researchcompliance/information-research-participants> for more information on your rights as a research participant.

SECTION ONE: PNMS KNOWLEDGE AND SUPPORT

(Kot el bliongel): klemedengei era PNMS ma ochodbechel)

Q1. How would you rate your level of knowledge of the Palau National Marine Sanctuary (PNMS)?

___ (0) No knowledge (**Skip to section 3**) (Diak (bora section 3))

___ (1) Little level of knowledge

___ (2) Medium level of knowledge

___ (3) High level of knowledge

___ (4) Extensive level of knowledge

Q2. Have you heard, seen, or read anything about PNMS, and/or participated in any outreach or education activities related to the PNMS?(Kemla rongesii, mesang, chuieuii

ma lechub e teloi era ngiidil cheldecheduch el kirel chisel ma lechub a beketeklel a ureor el kirel a tekoi era PNMS?)

___ (0) No (**Skip to section 3**) (Diak (bora section 3)

___ (1) Yes (mark yes or no in each that apply) (Choi, (ka molilt a ikel rokui el choblam ngarngii)

Source of information	YES (1)	NO (0)
Fact sheets/printed materials (brochures, posters, flyers) (babier el lolecholt era teletelel a PNMS)		
School outreach programs (Earth Day fairs, PICRC) (rengalek era skuul a merael louchais)		
Television (TV)		
Newspaper (Shimbung)		
Radio (Rasio)		
Community/town hall meeting (ongdibel era rebeluu)		
Symposiums/conferences/lectures (meklou el ongdibel)		
Internet/Social media or online sources (FB, Instagram, Google) (dengua ma lechub eng internet)		

Other (please specify) (bebil (ka beketeklii) _____ _____		
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Q3. If yes, what are the objectives of the PNMS? Meng ngerang aikel moktek er tial PNMS? Ka basech aikel rokui el moktek el modengei. [ENUMERATOR: DO NOT READ THE STATEMENT OF OBJECTIVES. ASK, RECORD, AND KEEP ASKING UNTIL THERE IS NO MORE ANSWER FROM THE PARTICIPANT; MARK YES IF THEY SAY ANY OF THE HIGHLIGHTED KEY WORDS].

Statements about PNMS's objectives (tekoi el kirel a moktek era PNMS)	YES (1)	NO (0)
1. Healthy ocean (ungil a klengar era daob)		
2. Healthy coral reef (ungil a klengar era kereker)		
3. Benefit Palau's economy (klungiolel a keisai era Belau)		
4. Benefit Palauan's livelihood (klungiolel a klengar era rechad)		
5. Increase marine resources (omdois a dikesel a daob)		
6. Maintain supply of nutritious and safe fish (seafood) for everyone (melemolem el ngarngii a ungil odoim el mora rokui el chad)		
7. Develop domestic pelagic fishing industry (mo mekedmokel a chelechad el kirel a omenged era ngeuaol)		

8. Support existing sustainable tourism (mo olngeseu a ikel ngarngii el tekoi era klekangkodang)		
9. Development that meets the needs of the present generation without compromising the needs of the future generation (sustainable development) (ng kedmekill el tekoi, el ngii a mo ruedem a ikel dousbech er chelchal telecheroll, e diak bo temall a ikel lousbech er sel merael mei el telecheroll)		
10. Others: (specify) (bebil (ka beketeklii) _____		

Q4. What is your level of support or opposition for the establishment of the 80% of Palau's Exclusive Economic Zone (EEZ) as a no-take zone prohibiting any type of fishing, entry and other exploitive actions (Ng uangerang a klungel a odbechem el kirel a osmerii a okai el basent era errengel a debel Belau, meng mo diak omenged er ngii)

- ____ (1) Strongly oppose
- ____ (2) Oppose
- ____ (3) Neither support nor oppose
- ____ (4) Support (**SKIP to Q6**)
- ____ (5) Strongly support (**SKIP to Q6**)
- ____ (75) I don't know (ng diak kudengei)

Q5. If you indicated "Strongly oppose" or "Oppose" please give 2 main reasons explaining your level of opposition. (a lsekum ke ngileltii a "**diak kuldubech**" ma lechub eng "**telkib loldubech**", ka mouchais a erul luchul e ke mililt a ikang)

1. _____
2. _____

Q6. Have you been negatively impacted by the PNMS?

- ____ (0) No
- ____ (1) Yes, specify: _____
- ____ (75) I don't know

Q7. Based on the list of needs below, please indicate whether you feel your family and community needs were taken into consideration during the planning and implementation of the PNMS?

	Family			Community		
	YES (1)	NO (0)	UNSURE	YES (1)	NO (0)	UNSURE
Food needs						
Economic needs						
Recreation needs						
Cultural needs						

Q8. State your level of agreement to the following statement: PNMS benefits all people of Palau equitably. (Ka mouchais era klisichel a uldesuem el kirel tial mo medung el tekoi: a PNMS a tabesul el klungiaol el mora rokui el chad era Belau)

- ____ (1) Strongly disagree (ng kmal diak kengei)
- ____ (2) Disagree (telkib el diak kengei)
- ____ (3) Neither (diak kengei ma diak lak kengei) (Skip to Q10)
- ____ (4) Agree (telkib el kongei) (Skip to Q10)
- ____ (5) Strongly agree (a kmal kongei)(Skip to Q10)

____ (75) Unable to assess (diak el sebechek el longer er tial tekoi)(Skip to Q10)

Q9. If you answered strongly disagree or disagree (a lsekum eng dimlak em kengei)ng sebechek loker el kmo: may I ask:

Who benefits the most? _____

(te ngerang a mo ngmai a klungiaol)

Who benefits the least? _____

(te ngerang a diak el ngai a klungiaol)

Who is or will be negatively impacted? _____

Q10. Please give one suggestion to the management of the PNMS so that it can provide equitable benefits to all people of Palau.

1. _____

Q11a. Have you heard about efforts to change the PNMS law from the original policy?

____ (0) No (skip to Section 2)

____ (1) Yes

Q11b. Do you support the efforts to change the PNMS law from the original policy?

____ (0) No (skip to Q11c)

____ (1) Yes (skip to Q11d)

____ (75) Unable to assess/I don't know (skip to Section 2)

Q11c. Why do you oppose efforts to change the PNMS law from the original policy?

Q11d. Why do you support efforts to change the PNMS law from the original policy?

SECTION TWO: PERCEIVED BENEFITS FROM PNMS

Q12. What is your level of agreement on the following?

(Ng uangerang a klungel a kengei er kau el kirel a ikal beldukel er iou?)

Statements (meldung el tekoi)	1 = strongly disagree	2 = disagree	3 = neither	4 = agree	5 = strongly agree	75 = Unable to assess/ don't know	77 = Not applicable
1. PNMS helps Palau attract quality and eco-friendly tourists. (a PNMS a olngeseu era Belau el mengesuseu era rungil e mengull era lukel a klengar el kangkodang)							
2. PNMS helps conserve reefs to stay healthy for dive and tourism industries. (a PNMS a olngeseu el korremelii a kereker meng melemolem el ungil kirel a tekoi era omelehelbakel ma klekangkodang)							
3. My household has a sustained, safe, and nutritious supply of seafood. (a delengcheklek a ngarngii a melemolem e keldmokol el roel el melai ungil el odoim era daob)							
4. Offshore pelagic fish is affordable food for my household. (a ngikel ngeuaol a beot lobechar elmo odoim era delengcheklam)							
5. Currently PNMS has no benefit for my household. (chelechal taem, ea PNMS a diak a klungiolel el mera delengcheklek)							

6. PNMS makes it difficult for my household to fish the way we used to. (a PNMS a rulii meng mo chetituokel era delengcheklek a oumenged el ua teletelam era ngaramong)							
7. Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans (a debel ngeuaol ma ikel ngara chelsel a diak el sal meklou a ututelel el ua kereker remel ma ikel ngara chelsel el meklou a ututelel el kirir a rechad era Belau)							
8. PNMS helps the reefs to stay strong to better protect the coastline and coastal communities. (a PNMS a olengeseu era kereker meng mo mesisiich, meng mo sebechel el sobecheklii a kederang ma ikel delengchokel el ngara kederang)							
9. Permanent closure or no-take from fisheries by PNMS helps Palau sustain and replenish its fisheries. (a omengesmerel ma lechub eng diak omeleiul a ngikel el loku a PNMS a ngoseuir a Belau meng melemolem el di ngarngii, e dirrek elmo betok a ngkelel)							
10. PNMS helps revive and strengthen our traditional connection to our environment and resources. (a PNMS a olengeseu el okiis e smisiich a uldelid el klechibelau er kid el kirel a lukel a klengar ma dikesed)							

11. PNMS helps maintain Palau's marine resources that are important to our sense of place and identity. (a PNMS olengeseu, meng melemolem el ngarngii a dikesel a debel Belau, el ngii a meklou a ultutelel el kirel a klechibelau er kid)							
12. PNMS management has a good process to involve communities in their decisions and activities. (a PNMS a ngarngii a ungil rael el lousbech er ngii el kidebii a beluu e ouchais a tekoi el mla metib ma ureor el loruul)							
13. PNMS benefits the next generations. (a PNMS a klungioliir ar merael el mei el telecheroll)							
14. PNMS creates the next generation of Palauan leadership to manage Palau's open ocean resources. (a PNMS a mekedbechii a reng era ngerachel, era re merael mei el telecheroll, el kirel a siungel a dikesel Belau era ngeuaol)							
15. Palau has the needed capacity to manage no-take and domestic fishing zones within the PNMS. (A Belau a ngarngii a ikel lousbech el duch elmo orrurt er sel mla mechesimer el basio me sel subed a dengara chei er ngii.							
16. PNMS makes the conservation efforts of Palau known internationally. (a PNMS a rulii ma urered era omengeriomel a bolodengalii a beluulechad)							
17. It is necessary to reexamine and revise the way the PNMS law is currently structured.							

18. Changing or revising the current PNMS law will bring prosperity and increased economic opportunity to Palau.							
19. Changing the current PNMS law will harm Palau's environment.							
20. Changing the current PNMS law will harm Palau's international conservation reputation/standing.							

SECTION THREE: SUSTAINABLE LIVELIHOOD AND FOOD SECURITY

Q13. In the pst 12 months, has your household participated in any of the following fishing activities? (A ikal merekong el truich ma erul buil, ng mlarngii a tal taem el tial delengchekliu a kora mlo ulsiseb era tang era ikal beldukel er iou el telengtengil a omenged?)

___ (1) Offshore pelagic fishing (omenged era ngeuaol) (if answered, go to Q14, but only show pelagics column)

___ (2) Reef fishing (omenged eremel) (if answered, go to Q14, show only reef fish column)

___ (3) Both (ngii lerung) (if answered, go to Q14, show both columns)

___ (4) My household does not participate in any type of fishing activities (**SKIP TO Q17**)

(Tial delengcheklam a diak lulsiseb era ngiidil telengtengil a tekoi era omenged)

Q14. In the past 12 months, how often did your household participate in pelagic and/or reef fishing? A ikal merekong el truich ma erul buil, ng mle uangerang a keldmel a delengchekliu el kirel a tekoi era omenged era ngeuaol ma lechub a omenged remel?

	Pelagic offshore fishing (omenged era ngeuaol)	Reef fishing (omenged eremel)
1. Rarely (ng kmal derstang)		

2. Once or twice a month(tang malechub eng erung era tal buil)		
3. About once a week (daitai el tang era tal sandei)		
4. 3-4 times a week(edei elmo eual taem ra tal sandei)		
5. Daily (bekel sils)		

Q15. Is reef and/or pelagic fishing for the following purposes? (Click yes or no for all that apply) A omenged remel ma lechub a ngeuaol, ng mora ikal beldukel er iou el moktek?

Purpose (moktek)	Pelagic offshore fish (omenged era ngeuaol)		Reef fishing (omenged eremel)	
Household consumption (ng di odoim)	(1) Yes	(0) No	(1) Yes	(0) No
Sale (ngoteruul) (if Yes, go to Q17)	(1) Yes	(0) No	(1) Yes	(0) No
Give away to friends and relatives (merous el mora sechelimiu ma rechedemiu)	(1) Yes	(0) No	(1) Yes	(0) No
Recreation (ng diomilil lomenged)	(1) Yes	(0) No	(1) Yes	(0) No
Other (specify) (bebil (ka beketeklii)	(1) Yes	(0) No	(1) Yes	(0) No

Q16. If you sell pelagic or reef fish, who do you sell it to and what is the average per pound price that you receive? (Click yes or no for all that apply)

Buyers	Pelagic fish		Pelagic price/lb	Reef fish		Reef fish price/lb
Fish market	(1) Yes	(0) No		(1) Yes	(0) No	
Supermarket	(1) Yes	(0) No		(1) Yes	(0) No	

Hotel Restaurant	(1) Yes	(0) No		(1) Yes	(0) No	
Restaurant	(1) Yes	(0) No		(1) Yes	(0) No	
Directly to community/individuals	(1) Yes	(0) No		(1) Yes	(0) No	
Belau Offshore Fisheries Incorporated (BOFI)	(1) Yes	(0) No		(1) Yes	(0) No	
Northern Reef Fisheries Cooperative (NRFC)	(1) Yes	(0) No		(1) Yes	(0) No	
Government (MOE, MOJ, MHHS, etc)	(1) Yes	(0) No		(1) Yes	(0) No	
Caterers	(1) Yes	(0) No		(1) Yes	(0) No	
Makit	(1) Yes	(0) No		(1) Yes	(0) No	
Others (specify): _____	(1) Yes	(0) No		(1) Yes	(0) No	

Q17. In the past 12 months, how often has pelagic offshore and/or reef fish been used by your household for customary and traditional events? A ikal merekong el teruich ma erul buil, ng uangerang a keldmel a delengchekliu el ousbech era ngikel ngeuaol ma lechub a klengikel el kirel a tekoi era mechsang ma lechub a siukang?

		Pelagic offshore fishing (omenged era ngeuaol)	Reef fishing (omenged eremel)
0	Never (ng dimlak a tang)		
1	Once or twice a year		
2	Every other month		
3	Once or twice a month		
4	Always (di blechoel)		

Q18. Between pelagic offshore fish and reef fish, which does your household prefer to eat? Tial delengchekliu ng sorir el mengang a ngikel ngeuaol ma lechub eng klengikel?

- _____ (1) Pelagic Offshore fish (ngikel ngeuaol) (go to Q20, show only pelagic offshore column)
- _____ (2) Reef fish (klengikel) (go to Q20, show only reef fish column)
- _____ (3) Neither (My household does not eat reef or pelagic fish) **[SKIP TO Q23]**
- _____ (4) Both (ngii lerung) (go to Q20, show both columns)

Q19. Please explain why your household prefers eating pelagic offshore or reef fish or both as answered in Q5. (Click yes or no for all that apply.) Ka mouchais el kmo, tial delengchekliu, ngera uchul meng sorir el mengang a ngikel ngeuaol ma lechub eng klengikel ma lechub eng ngii lerung elua isel mulenger er ngii el ngara Q5.

Reason (uchul eng sorir)	Pelagic offshore fish (ngikel ngeuaol)		Reef fish (klengikel)	
Price (affordability) (ng cheral)	(1) Yes	(0) No	(1) Yes	(0) No
Availability (ng blechoel el ngarngii)	(1) Yes	(0) No	(1) Yes	(0) No
Taste preference (ng telemtemul)	(1) Yes	(0) No	(1) Yes	(0) No
High quality food (ngungil el odoim)	(1) Yes	(0) No	(1) Yes	(0) No
Knowledge and preparation of cooking (ng klaodengei era omerellel)	(1) Yes	(0) No	(1) Yes	(0) No
Variety (bek el bedengel)	(1) Yes	(0) No	(1) Yes	(0) No
Ease of consumption	(1) Yes	(0) No	(1) Yes	(0) No
More meat/feeds more people	(1) Yes	(0) No	(1) Yes	(0) No
Others (specify) _____	(1) Yes	(0) No	(1) Yes	(0) No

Q20. Where do you get fish for food in your household? (Check all that apply) (Ko melai a odimiu el ngikel ngeuaol er ker?)

Source of Fish	Pelagic offshore fish (ngikel ngeuaol)		Reef fish (klengikel)	
Caught by myself or someone else in our household (ke chilteklii malechub eng chad era blimiu a chilteklii?)	(1) Yes	(0) No	(1) Yes	(0) No

Bought from fish market (local) (ke mlecherar era makit /stoang)	(1) Yes	(0) No	(1) Yes	(0) No
Bought from fish market (imported) (ke mlecherar era makit /stoang)	(1) Yes	(0) No	(1) Yes	(0) No
Bought from supermarket (local)	(1) Yes	(0) No	(1) Yes	(0) No
Bought from supermarket (imported)	(1) Yes	(0) No	(1) Yes	(0) No
Bought from Belau Offshore Fisheries Incorporated (BOFI)	(1) Yes	(0) No	(1) Yes	(0) No
Bought from Northern Reef Fisheries Cooperative (NRFC)	(1) Yes	(0) No	(1) Yes	(0) No
Bought from a neighbor, friend, or other person (not from market or store) (ke mlecherar era tonari, sechelim ma lechub eng tara chad (ng diak el makit/stoang)	(1) Yes	(0) No	(1) Yes	(0) No
Restaurant (ke di mengang era blil a blengur)	(1) Yes	(0) No	(1) Yes	(0) No
Given by family or friends (ng milskemiu a rechedemiu ma lechub eng sechelimiu)	(1) Yes	(0) No	(1) Yes	(0) No
Others (please specify _____)				

Q21. If you buy pelagic or reef fish, what is the average price you pay per pound and do you think that is affordable?

Buy Pelagic offshore fish? (ngikel ngeuaol)		Pelagic fish Price/lb	Pelagic fish price affordable?		Buy Reef fish?		Reef fish price/lb	Reef fish price affordable?	
(1) Yes	(0) No		(1) Yes	(0) No	(1) Yes	(0) No		(1) Yes	(0) No

Q22. If you buy offshore pelagic fish from the market or store, is it available for sale when you want it? A lsekum ko mechar a ngikel ngeuaol era makit ma lechub eng stoang, el mo odoim, ng sebechem el blechoel nguu sel bol soam?

- _____ (0) Never (diak a tang)
- _____ (1) Rarely (kmal derstang)
- _____ (2) Sometimes (bebil era taem)
- _____ (3) Often (mekudem)
- _____ (4) Always (blechoel)

_____ (5) I do not buy offshore pelagic fish (ng diak kumechar a ngikel ngeuaol

Q23. On average in the past 12 months, how often does your household eat the following odoim? Name all categories. A ikaI merekong el truich ma erul buil, ng daitai el uangerang a keldmel tial delengchekliu era omengelel a ikaI beldukel er iou el odoim? Ka basech a ikaI rokui el bedengel.

	Very rarely/ Never (kmal dersta ng ma lak)	Once or twice a month (ta ng malechub eng erung era tal buil)	About once a week (daitai el tang era tal sandei)	3-4 times a week (edei elmo eual taem ra tal sansdei)	Daily (bekel sils)	I don't know (ng diak kudengei)
Fresh Tuna (beches el tuna)						
Processed tuna (e.g. smoked, flakes, dried) (mla mekedmokeI el tuna (eltuul,nglabek, ulekdiIrt)						
Fresh non-tuna pelagic offshore fish (e.g. mahi mahi, wahoo, marlin) (beches el ngikel ngeuaol el diak el tuna)						
Processed non-tuna (e.g. smoked, flakes, dried) (mla mekedmokeI eldiak el ngikel ngeuaol (e.g. eltuul, nglabek, ulekdiIrt)						
Imported fish (salmon, basa, tilapia)						
Fresh reef fish (beches el klengikel)						
Processed reef fish (e.g. smoked, flakes, dried) (mla mekedmokeI el klengikel (eltuul,nglabek, ulekdiIrt)						
Canned meat (canned tuna, SPAM etc) (kansume)						

Q24a. In the past 12 months, have impacts from COVID-19 altered your fishing behavior?

___(1) No (skip to Section Four)

___(2) Yes

Q24b. How did impacts from COVID-19 altered your fishing behavior? (check all that apply)

	YES (1)	NO (0)
Went fishing less		
Went fishing more		
Changed what species I targeted		
Changed from offshore to nearshore		
Changed from nearshore to shore		
Changed what gear I used		
Other (please specify): _____		

Section Four: Socio-demographics. Q25. (Telengtengil a delengchokl)

A. Household ID	B. State	C. Gender	D. Age (in years) Ng tela rekim?	E. Highest level of formal education (dauch era skuul)	F. What are the top 3 sources of income for your household? (ngera ikel keldei el meklou luchul a kerrekerngem)	G. Citizenship (beluam)	H. How many people live in your household (including yourself)? (Kom telal chad l kiei rtial deleng chokl luldumukl rekau?)	I. What is the annual income level of your household? (ng uanger ang a klungel a kerrekerngel tial deleng chokl ra tal rak)
	1=Kayangel 2=Ngarchelong 3=Ngaraard 4=Ngiwal 5=Melekeok 6=Ngchesar 7=Airai 8=Aimeliik 9=Ngatpang 10=Ngere mlengui 11=Ngard mau 12=Koror 13=Peleliu 14=Angaur 15=Sonso rol	1= 'Male' 2= 'Female'	1=18-24 years 2=25-44 years 3=45-64 years 4=65-79 years 5=80 and over 999=Refuse to answer	1= Elementary 2= High school 3= Vocational school/training 4= Some College (2-year college degree) 5= University (4 year, Bachelor's degree)	1=Handicraft (klalou el meruul era chim) 2a=Pelagic fishing (omenged era ngeuaol) 2b=Reef fishing (omenged remel) 3=Farming and livestock) (sers ma okerulel a charm) 4=Family-owned business (siobai era telungalek) 5=Remittances (odureklel a udoud) 6=Land or house lease (chutem ma lechub eng blai el lengiil) 7=Government work	1=Palau 2=Micronesia 3=Philippines 4=Bangladesh 5=USA 6=Japan 7=Taiwan 8=China 9=Korea 10=Other (specify)		1= Less than \$5,000 2 = \$5,000-9,999 3 = \$10,000-19,999 4 = \$20,000-29,999 5 = \$30,000-39,999 6 = \$40,000-49,999 7 = \$50,000

	16=Hatohebei			6= Master's degree and/or PhD 7=None	(ureor era Amt) 8=Pension/social security (pension ma ss) 9=Tourism-related businesses and services (kleangkodang ma rokui el telengtengil) 10=other (specify) (bebil, (ka beketeklii) _____			0-59,999 8 = more than \$60,000 999 = refused to answer
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